Toward Principled Threshold Selection for Blink Region Detection in Biological Signals: A Cross-Dataset Benchmark with LLM-Guided Candidate Derivation

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Abstract

Blink detection from electroencephalography (EEG) serves two complementary purposes: blinks are a major physiological contaminant that must be identified for artifact handling, and they are also informative behavioral markers linked to arousal, fatigue, cognitive load, and operational performance. Despite substantial progress in blink detection, a practically important setting remains comparatively under-addressed: epoch-structured recordings in which a long continuous signal is divided into non-overlapping analysis windows. This setting is the norm in driving neuroscience and other neuroergonomic paradigms, yet existing detectors—including BLINKER and MNE-based amplitude routines—operate on concatenated signals and implicitly pool blink-containing and blink-free epochs when estimating thresholds. We show that this pooling may bias threshold placement and alter the precision-recall balance. We propose a three-stage epoch-aware pipeline. Stage A screens epochs by their peak-to-peak amplitude via an autoreject-inspired cross-validation criterion, yielding a set of suspicious (likely blink-containing) epochs. Stage B estimates a robust sample-level threshold from the suspicious epoch pool using the median and median absolute deviation (MAD), concentrating estimation on signal segments enriched for blink-like excursions. Stage C applies the threshold to extract blink regions from the suspicious epochs and maps detections back to epoch-local coordinates. A fallback mechanism ensures stability when too few suspicious epochs are available. We evaluate four conditions—BLINKER-concat, MNE-annot, Proposed-Mean, and Proposed-Med—on two naturalistic driving-EEG corpora, Raja and Cao2018, under epoch durations of 30, 40, and 60 seconds, with 30 seconds selected as the primary setting. Statistical comparisons use macro-averaged event-level F_1 with Wilcoxon signed-rank tests. Additional analyses examine stability across boundary tolerances. Results demonstrate that epoch-aware threshold estimation consistently outperforms naive concatenation, that the robust MAD-based estimator is preferable to a mean-based alternative, and that the robust median configuration (Proposed-Med) achieved the best macro-averaged event-level F_1 (pooled 0.867) and significantly outperformed the baselines and the mean-based variant (Wilcoxon, Bonferroni).

1 Introduction

Blinking is a physiological process that involves the closure and reopening of the eyelids [1]. Its primary functions are to protect the ocular surface, maintain corneal moisture, and support visual comfort [1]. Beyond these physiological functions, blink behaviour has been increasingly recognised as an important indicator in fatigue-driving research [2, 3]. Variations in blink-related features, such as blink frequency, blink duration, eye-closure duration, and blink patterns, can reflect changes in alertness and cognitive state [2, 4]. As fatigue increases, drivers may exhibit slower or prolonged blinks, which can indicate reduced vigilance and a higher risk of impaired driving performance [2]. Therefore, blink behaviour provides a useful and non-invasive marker for monitoring driver fatigue [5, 3].

In the neuroscience study of fatigue driving, blink activity can be captured through external methods, such as camera-based eye monitoring, or through electrophysiological recordings, particularly electroencephalography and electrooculography [3, 5]. Although EEG is mainly used to measure brain activity, blink-related components are often present in EEG signals because eyelid movements and eye movements generate strong electrical potentials [6, 7]. These ocular activities are especially prominent in frontal EEG electrodes due to their close proximity to the eyes [6, 8]. As a result, blink information can be extracted from EEG recordings, either by analysing the ocular components embedded in the EEG signal or by combining EEG with EOG-based measurements [9, 5]. This makes EEG/EOG recordings valuable not only for studying neural activity during fatigue but also for identifying blink-related markers associated with drowsiness [10, 2].

It is important to distinguish between two major perspectives on blink activity in the existing literature [6, 3]. In many EEG studies, blink-related activity is treated as an artifact because it contaminates neural signals and may interfere with the interpretation of brain activity [6, 7]. In such cases, blink components are commonly removed during EEG preprocessing [6, 11]. However, in fatigue-driving studies, blink activity can also be treated as meaningful information rather than noise [3, 5]. From this perspective, blink events are identified, extracted, and analysed as potential indicators of driver fatigue [10, 5]. Blink detection methods can generally be classified into machine-learning-based, morphology-based, and thresholding-based approaches [9, 12]. Machine learning methods rely on trained models to classify blink events, while morphology-based methods identify blinks based on their characteristic waveform shape [13, 14]. Thresholding-based methods, in contrast, detect candidate blink events when the signal amplitude exceeds a certain threshold value, as blink activity typically produces large-amplitude deflections, especially in frontal channels [15, 8].

Thresholding is one of the most straightforward and widely used approaches for detecting blink activity from EEG or EOG signals [16, 15]. The simplicity of thresholding makes it computationally efficient and easy to implement, particularly in real-time fatigue-detection systems [16, 17]. In addition, thresholding is often used as an initial detection step in more complex pipelines, where the detected candidates are further refined using morphological rules or machine-learning classifiers [14, 18]. However, selecting an appropriate threshold value is not trivial [19, 20]. Blink amplitude can vary across participants, electrode locations, recording devices, signal quality, and experimental conditions [19, 20]. A threshold that is too low may produce false detections due to noise or other artifacts, whereas a threshold that is too high may miss genuine blink events [16, 17]. Therefore, threshold selection remains a critical consideration in the reliable detection of blink activity for fatigue-driving research [10, 5].

In many fatigue-driving studies, EEG recordings are segmented into epochs that correspond to specific task events or analysis windows [21]. This epoch-structured setting introduces additional challenges for threshold-based blink detection [16, 17]. When thresholds are estimated from the entire recording, they may be biased by the presence of clean epochs that contain few or no blinks, which may distort threshold estimates and suppress blink detection in epochs where blinks are more prevalent [11, 19]. As a result, naive application of thresholding in an epoch-structured context may risk under-detection of blinks, which undermines the reliability of fatigue assessments based on blink metrics [10, 5]. Addressing this challenge requires developing blink detection methods that explicitly account for the epoch structure and can adapt threshold estimation to the specific characteristics of each epoch or subset of epochs, rather than relying on a global threshold derived from the entire recording [19, 9].

This paper makes several contributions to the field of blink detection in fatigue-driving research. First, we formalise the problem of blink detection in the epoch-structured setting and demonstrate the limitations of naive thresholding approaches that do not account for the epoch structure. Second, we propose a three-stage pipeline for blink detection that is, to our knowledge, among the first to include epoch screening, sample-level threshold estimation, and blink-region extraction, which explicitly handles the challenges of the epoch-structured setting. Finally, we provide a comprehensive benchmark of competing methods across 30, 40, and 60 s epoch durations and a boundary-tolerance sensitivity analysis, all evaluated on two naturalistic driving-EEG corpora, Raja and Cao2018, under an event-level overlap criterion with macro-averaged F_1 and Wilcoxon signed-rank tests [21, 22].

2 Related Work

The classical literature established the difficulty of ocular contamination and motivated automated statistical detection directly from EEG. Early work emphasized artifact removal and statistical screening rather than blink-region benchmarking per se, but it already showed that extreme-value behavior, channel context, and morphology matter [6, 11]. This historical backdrop explains why simple one-value global thresholds can be attractive, yet brittle.

A first important stream comprises single-channel or few-channel threshold-centered detectors. Chang et al. [15] detect blink artifacts from a single prefrontal channel using digital filtering and rule-based decisions. Tran et al. [16] present an explicitly thresholding-based EEG blink detector and compute a minimum amplitude threshold from transformed signal magnitude and signal standard deviation. Zhang et al. [12] move toward real-time frontal EEG detection with short windows and decision logic around potential blink boundaries. Wang et al. [20] extend thresholding to harder clinical settings by optimizing multidimensional features in the presence of epileptiform discharges, and later work based on higher-order cumulants continues the move toward short-segment single-channel detection [23].

A second stream contains hybrid thresholding pipelines. BLINKER first identifies candidate intervals with a mean-based standard-deviation threshold and then rejects implausible candidates using robust amplitude-distribution criteria derived from the median absolute deviation [8]. Cao et al. [24] use cascaded thresholding over amplitude, amplitude displacement, and cross-channel correlation before applying a Gaussian mixture model. Agarwal et al. [9] self-learn user-specific profiles in an unsupervised manner, while Valderrama et al. [25] use iterative template matching and automatic threshold/template estimation for suppression in single-channel recordings. Guttmann et al. [19] further emphasize adaptation to inter- and intra-subject variability, underscoring that fixed thresholds are often not enough.

A growing stream of blink and ocular-artifact research replaces hand-tuned thresholds with data-driven models. Convolutional neural networks have been used both to detect eye-blink artifacts and to remove them from EEG [26, 27], while transformer- and video-based architectures have pushed blink-detection accuracy further [28, 29]. Autoencoder and segmentation-denoising networks also target single-channel artifact removal [30, 31], and recent reviews catalogue the broader shift from conventional thresholding toward deep learning for physiological-artifact handling, including in wearable EEG [32, 33]. In driving and fatigue monitoring, machine-learning and CNN models detect drowsiness from blink and eye-state cues [34, 35]. These data-driven detectors can be accurate, but they typically require labelled training data, more computation, and offer less transparency than a single interpretable threshold. This contrast motivates studying the lightweight thresholding stage itself, which frequently precedes or complements such pipelines, and in particular the epoch-structure estimation bias that this paper addresses.

3 Method

3.1 Problem Definition

Let a blink annotation be an interval $g_i = [s_i, e_i]$ and a detector output be an interval $d_j = [\hat{s}_j, \hat{e}_j]$. A detected region is counted as a true positive if it overlaps at least one unmatched ground-truth region. Unmatched detections are false positives; unmatched ground-truth regions are false negatives. This event-level treatment follows the logic of interval-based detection rather than pointwise accuracy, which is appropriate because blink boundaries are uncertain, annotation granularity varies, and non-event samples dominate the timeline [36, 37].

3.2 Datasets

Two naturalistic driving-EEG corpora were evaluated: Raja and Cao2018. The Raja corpus comprised 46 complete sessions recorded with a 128-channel EGI system (ANT Neuro), referenced to CPz under a modified international 10-20 layout, during a simulated lane-keeping driving task; ground-truth blink annotations were derived from human EAR/EOG labelling. The Cao2018 corpus was the sustained-attention driving corpus reported by [21], in which 32-channel EEG was recorded during an approximately 90-minute lane-keeping task; 58 complete sessions were used, and 4 non-complete sessions were excluded. Because the published preprocessed version of Cao2018 removed eye blinks, the raw recordings were used and paired with external blink-region annotations.

3.3 Preprocessing

All detectors used identical preprocessing, consisting of a 1–20 Hz band-pass filter, no re-sampling, and EEG-channel selection. The channel scope was dataset specific: for Raja, the analysis was restricted to a minimal frontal montage comprising three channels (E3, E9, and E22), whereas for Cao2018, all 32 EEG channels were provided to the pipeline and the best channel was selected separately for each session. Thus, Raja channel selection was confined to frontal sites by construction, while Cao2018 channel selection was performed over the full montage.

3.4 Epoch Segmentation and Detection Pipeline

Processing pipeline overview. The proposed detector is an epoch-aware, threshold-based pipeline that transforms a continuous multichannel EEG recording into a set of event-level blink regions through a fixed sequence of seven operations. Figure 1 summarises this sequence as a flow chart, from the raw recording through preprocessing, epoching, the three detection stages (A, B, and C), and the final event-level evaluation. The same pipeline is applied identically across both datasets and across every detector configuration, so that any difference in performance is attributable to the detector under test rather than to incidental differences in preprocessing or scoring.

Step-by-step description. First, the continuous recording is downsampled to a common sampling grid of 100 Hz (`RESAMPLE_RATE`). Standardising the sampling rate makes the amplitude thresholds and the minimum-duration gate comparable across recordings acquired on different hardware, and reduces the per-sample computational cost of the downstream stages without discarding any information needed to resolve a blink, whose spectral content lies well below the new Nyquist frequency. Second, a zero-phase band-pass filter from 1.0 Hz (`FILTER_LOW`) to 20.0 Hz (`FILTER_HIGH`) is applied; this legacy BLINKER-style band removes slow drifts and DC offsets at the low end and high-frequency muscle and line noise at the high end, isolating the low-frequency, high-amplitude deflection that characterises an eye blink. Third, each filtered recording is split into non-overlapping, fixed-length epochs of 30 s (`EPOCH_DURATION_S`); epochs whose health metric falls below the acceptance criterion are dropped so that grossly corrupted segments do not distort the data-driven thresholds estimated later. Epoching is what makes the thresholds *adaptive*: rather than deriving a single level for the entire session, the pipeline treats each short epoch as a local statistical context.

In Stage A, a per-channel peak-to-peak (PTP) amplitude statistic is computed for every epoch, and an *autoreject*-based procedure estimates a channel-specific PTP threshold from the data. An aggregation rule over a chosen channel-selection group, by default flagging an epoch whenever *any* channel in the group exceeds its threshold, marks the epoch as "suspicious" (blink-heavy). The output of Stage A is therefore a set $\mathcal{B}$ of epochs that are enriched for blink-related activity, which restricts the subsequent amplitude analysis to the segments where blinks are most likely to occur. In Stage B, a robust sample-level amplitude threshold is estimated using only the samples pooled from the flagged epochs, as $\theta = \xi + k\,(1.4826 \cdot \mathrm{MAD})$, where the centre ξ is the *median* in the primary configuration (or the *mean* in the contrast configuration), MAD is the median absolute deviation, the factor 1.4826 rescales the MAD to a standard-deviation-consistent dispersion, and k is the sensitivity multiplier. Estimating the threshold from the flagged epochs concentrates the calculation on blink-enriched signal, while the median-plus-MAD construction keeps the estimate stable

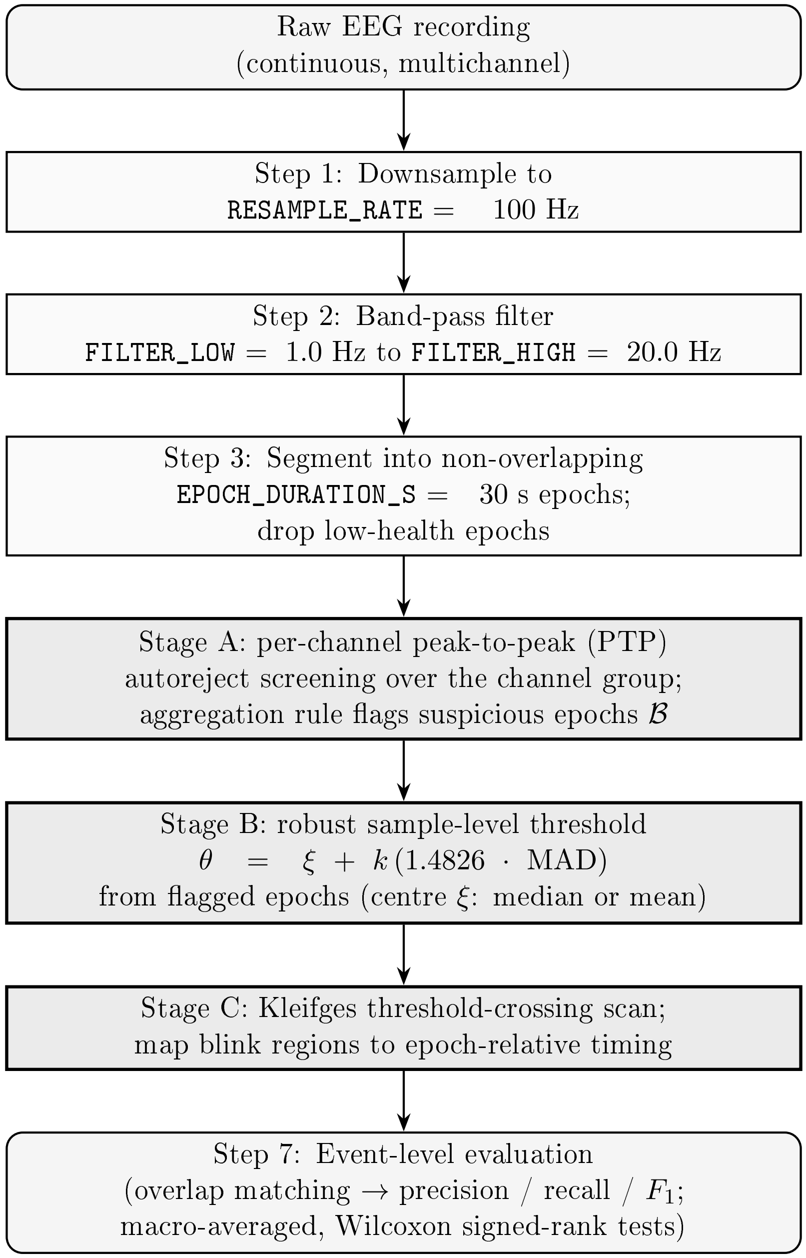

Figure 1: Per-epoch blink-detection pipeline. The continuous recording is downsampled and band-pass filtered, segmented into fixed-length epochs with low-health epochs removed, and then passed through the three detection stages: Stage A screens suspicious (blink-heavy) epochs by peak-to-peak amplitude, Stage B estimates a robust sample-level amplitude threshold from those epochs, and Stage C localises blink regions by scanning threshold crossings. Detections are finally scored against the ground truth at the event level.

in the presence of isolated extreme samples. If too few epochs are flagged, the estimate falls back to the full valid-epoch set so that the threshold is never derived from an unreliably small sample. In Stage C, the threshold is scanned across the signal in the manner of the Kleifges crossing detector: maximal contiguous runs of supra-threshold samples define candidate blink regions, brief crossings shorter than a minimum-duration gate are discarded, and the accepted regions are mapped from their concatenated coordinates back to epoch-relative onset and offset times. Finally, the detected regions are compared with the ground-truth annotations at the *event* level using overlap-based matching, yielding precision, recall, and F_1; these are macro-averaged across sessions and compared between configurations with Wilcoxon signed-rank tests, as detailed in Section 3.1. Recordings were segmented into fixed-length epochs, with 30 s used as the primary epoch length and 10, 20, 40, and 60 s additionally evaluated in an epoch-sensitivity sweep. The proposed detector was implemented as a three-stage epoch-aware threshold pipeline. First, suspicious/bad epochs were screened using an autoreject-based procedure with a fixed random state and a requirement that at least one epoch be flagged. Second, a sample-level amplitude threshold was estimated from the screened epochs using a robust centre and a standard-deviation multiplier of 3.5. Third, threshold crossings were expanded into blink regions. The detailed equations for each stage were given in the following subsections rather than duplicated in this overview.

3.5 Selection of Bad Epochs

The first stage of the proposed pipeline performs epoch-level screening using an autoreject-inspired threshold selection procedure. Its purpose is not to estimate the onset or offset of a blink, but to identify epochs whose amplitudes are sufficiently abnormal to warrant subsequent sample-level localization. Let $X \in \mathbb{R}^{N\times C\times T}$ denote the epoched multichannel signal after preprocessing, where N is the number of valid epochs, C is the number of channels, and T is the number of samples per epoch. The sample at epoch $i \in \mathcal{E} = \{1, \ldots, N\}$, channel $c \in \mathcal{C} = \{1, \ldots, C\}$, and within-epoch time index $t \in \{1, \ldots, T\}$ is denoted by $x_{i,c,t}$.

For each epoch and channel, an amplitude-range statistic is computed using the peak-to-peak value

$$a_{i,c} = \max_{1\le t\le T} x_{i,c,t} - \min_{1\le t\le T} x_{i,c,t}.$$

Large values of $a_{i,c}$ indicate that the corresponding epoch contains a high-amplitude excursion in channel c, a pattern consistent with ocular or other transient artifacts. Rather than imposing a fixed rejection level, the pipeline estimates a channel-specific threshold $\tau_c^\star$ from the data. Candidate thresholds are taken from the empirical range of peak-to-peak amplitudes for that channel, denoted $\mathcal{Q}_c = \{a_{i,c} : i \in \mathcal{E}\}$, and are evaluated by a cross-validation criterion.

Specifically, let $\{(\mathcal{T}_k, \mathcal{V}_k)\}_{k=1}^{K}$ be cross-validation splits of the epoch indices into training sets $\mathcal{T}_k$ and validation sets $\mathcal{V}_k$. For a candidate threshold $\tau \in \mathcal{Q}_c$, the training epochs retained for channel c in fold k are

$$\mathcal{G}_{k,c}(\tau) = \{i \in \mathcal{T}_k : a_{i,c} \le \tau\}.$$

These retained epochs define a threshold-dependent training template

$$\bar{x}_{k,c}^{(\tau)}(t) = \frac{1}{|\mathcal{G}_{k,c}(\tau)|} \sum_{i \in \mathcal{G}_{k,c}(\tau)} x_{i,c,t},$$

where candidates yielding an empty retained set are treated as inadmissible. The corresponding validation reference is computed as the pointwise median across the validation epochs,

$$\tilde{x}_{k,c}(t) = \mathrm{median}_{i \in \mathcal{V}_k} \, x_{i,c,t}.$$

The use of a median validation reference makes the criterion less sensitive to large validation-set artifacts. The validation loss for channel c is then defined as

$$\mathcal{J}_c(\tau) = \frac{1}{K} \sum_{k=1}^{K} \left[\frac{1}{T} \sum_{t=1}^{T} \left\{ \bar{x}_{k,c}^{(\tau)}(t) - \tilde{x}_{k,c}(t) \right\}^2 \right]^{1/2}.$$

The selected channel threshold is the candidate that minimizes the average validation discrepancy,

$$\tau_c^\star = \arg\min_{\tau \in \mathcal{Q}_c} \mathcal{J}_c(\tau).$$

In practice, this objective may be searched by sequential Bayesian optimization or by a random search over candidate peak-to-peak thresholds; both choices retain the same validation objective and rejection rule.

After the thresholds $\{\tau_c^\star\}_{c=1}^{C}$ have been estimated, each epoch receives a channel-level artifact indicator

$$z_{i,c} = \mathbf{1}\{a_{i,c} > \tau_c^\star\},$$

where $\mathbf{1}\{\cdot\}$ is the indicator function. The present pipeline uses the union of the channel-level decisions to form the epoch-level screening mask,

$$b_i = \mathbf{1}\left\{ \sum_{c=1}^{C} z_{i,c} \geq 1 \right\} = \mathbf{1}\left\{ \exists c \in \mathcal{C} : a_{i,c} > \tau_c^\star \right\}.$$

The set of suspicious epochs is therefore

$$\mathcal{B} = \{ i \in \mathcal{E} : b_i = 1 \}.$$

These epochs are interpreted as likely blink-contaminated or artifact-heavy segments. They provide the restricted data subset from which the next stage estimates sample-level blink-region thresholds.

3.6 Computation of Sample-Level Blink-Region Threshold Candidates

The second stage converts the epoch-level artifact decision into a sample-level thresholding problem. The autoreject-style screening in the previous stage identifies a set of suspicious epochs $\mathcal{B}$, but its thresholds $\tau_c^\star$ are defined on peak-to-peak amplitudes over whole epochs.

Such thresholds are therefore appropriate for deciding whether an epoch is abnormal, but they are not directly interpretable as sample-wise blink crossing levels. Stage B estimates a separate channel-specific threshold in the amplitude domain of the time series itself.

Let $\mathcal{S}$ denote the set of epochs used for estimating the sample-level threshold. In the primary analysis, $\mathcal{S} = \mathcal{B}$, so that the estimate is driven by epochs already identified as likely to contain blink-related activity. If too few suspicious epochs are available, the method falls back to the full valid epoch set $\mathcal{E}$ to avoid estimating a threshold from an insufficient sample:

$$\mathcal{S} = \begin{cases} \mathcal{B}, & |\mathcal{B}| \geq n_{\min}, \\ \mathcal{E}, & |\mathcal{B}| < n_{\min}, \end{cases}$$

where $n_{\min}$ is the minimum number of screened epochs required for a stable estimate.

For each channel c, all samples from the selected epochs are pooled into a one-dimensional empirical sample set

$$\mathcal{Y}_c = \{x_{i,c,t} : i \in \mathcal{S},\ 1 \leq t \leq T\}.$$

This pooling step marks the methodological transition from epoch-level rejection to sample-level localization: the quantities used in Stage B are no longer peak-to-peak ranges $a_{i,c}$, but the individual signal amplitudes $x_{i,c,t}$ that will later be thresholded to form candidate blink regions.

The sample-level threshold is estimated using a robust location-plus-dispersion form. Let ξ_c be the channel-specific center of the sample distribution. The default choice is the median,

$$\xi_c = \mathrm{median}_{y \in \mathcal{Y}_c}\, y,$$

although the arithmetic mean may be substituted when a mean-centered threshold is required for comparison. Dispersion is estimated by the median absolute deviation (MAD),

$$\mathrm{MAD}_c = \mathrm{median}_{y \in \mathcal{Y}_c} \left| y - \mathrm{median}_{u \in \mathcal{Y}_c}\, u \right|,$$

and is rescaled to a standard-deviation-compatible quantity by

$$\sigma_c^{\mathrm{rob}} = 1.4826\ \mathrm{MAD}_c\,.$$

The constant 1.4826 is the usual normal-consistency factor for the MAD. For a sensitivity parameter $k > 0$, the resulting sample-level blink-region threshold candidate for channel c is

$$\theta_c = \xi_c + k\sigma_c^{\mathrm{rob}}.$$

The set

$$\Theta = \{\theta_c : c \in \mathcal{C}\}$$

constitutes the channel-wise threshold candidates passed to the blink-region extraction stage.

This construction deliberately separates the statistical role of the two thresholds in the pipeline. The quantities $\tau_c^\star$ from Stage A are epoch-rejection thresholds defined on peak-to-peak amplitudes, whereas θ_c is a sample-level amplitude threshold defined on the time-series samples themselves. Estimating θ_c from $\mathcal{S}$ concentrates the calculation on signal segments enriched for blink-like excursions, while the robust center and MAD-scaled dispersion reduce the influence of isolated extreme peaks within those same segments.

3.7 Extraction of Blink Regions

The final stage applies the sample-level thresholds $\Theta = \{\theta_c\}_{c=1}^{C}$ to obtain channel-wise blink regions. For each channel c, the epochs selected for localization are ordered as $\mathcal{S} = \{i_1, \ldots, i_R\}$, where $R = |\mathcal{S}|$. Their samples are concatenated into a one-dimensional signal

$$u_c[n] = x_{i_r,c,t}, \qquad n = (r-1)T + t, \quad r \in \{1, \ldots, R\}, \quad t \in \{1, \ldots, T\}.$$

Thus $u_c \in \mathbb{R}^{RT}$ is the channel-specific signal on which blink onsets and offsets are estimated. If the analysis is run in the fallback mode described above, $\mathcal{S}$ is the full valid epoch set; otherwise it is the set of epochs identified as suspicious in Stage A.

Candidate blink samples are defined by a threshold-crossing indicator

$$h_c[n] = \mathbf{1}\{u_c[n] > \theta_c\}, \qquad n = 1, \ldots, RT.$$

After any required polarity normalization, $h_c[n] = 1$ indicates that the sample amplitude exceeds the channel-specific blink-region threshold. Blink regions are then defined as maximal contiguous runs of supra-threshold samples. A candidate interval for channel c is written as the half-open index range $[s_{c,r}, e_{c,r})$, where

$$h_c[n] = 1 \quad \text{for all } n = s_{c,r}, \ldots, e_{c,r} - 1,$$

with either $s_{c,r} = 1$ or $h_c[s_{c,r} - 1] = 0$, and either $e_{c,r} = RT + 1$ or $h_c[e_{c,r}] = 0$. Equivalently, the onset $s_{c,r}$ is the first sample in a run that crosses above θ_c, and the offset $e_{c,r}$ is the first subsequent sample at which the signal has returned to the sub-threshold state.

To suppress brief threshold crossings that are unlikely to represent complete blinks, each candidate interval is required to exceed a minimum duration. Let f_s be the sampling frequency and let $\ell_{\min}$ be the minimum blink duration in seconds. The corresponding sample length is

$$L_{\min} = \ell_{\min} f_s.$$

The accepted blink-region set for channel c is

$$\mathcal{R}_c = \left\{(s_{c,r}, e_{c,r}) : e_{c,r} - s_{c,r} > L_{\min}\right\}.$$

This duration gate is applied independently for each channel. Consecutive supra-threshold samples are merged into the same region by construction, while two runs separated by at least one sub-threshold sample are treated as distinct candidate blink regions.

Finally, the accepted intervals are mapped from concatenated coordinates back to epoch-local coordinates using the known epoch boundaries. If an accepted onset $s_{c,r}$ lies in the concatenated segment corresponding to epoch i_q, its within-epoch onset sample is

$$\hat{s}_{c,r} = s_{c,r} - (q-1)T,$$

and its onset time is $(\hat{s}_{c,r} - 1)/f_s$ seconds relative to the beginning of that epoch. The interval duration is

$$\hat{d}_{c,r} = \frac{e_{c,r} - s_{c,r}}{f_s}.$$

The final output of this stage is therefore a channel-indexed collection of blink regions, each represented by an epoch identifier, a channel label, an onset index or time, and an offset or duration. The extraction remains channel-wise throughout: the threshold θ_c estimated for channel c is applied only to that channel's signal, and the resulting regions $\mathcal{R}_c$ are retained as channel-specific blink candidates.

3.8 Detector Configurations

Four detector configurations were compared under identical preprocessing and epoching. The proposed methods were Proposed-Med, which implemented the three-stage pipeline with a median centre, and Proposed-Mean, which used a mean centre. The conventional baselines were BLINKER-concat and MNE-annot.

3.9 Experiment Overview

The experiments follow a deliberate logical progression. Experiment 1 first selects the best Stage-A channel-selection configuration; Experiment 2 then benchmarks that chosen configuration against competing detectors; and Experiments 3–8 stress-test and characterise the resulting pipeline along orthogonal axes (epoch duration, boundary tolerance, fallback sensitivity, blink morphology, epoch-health exclusion, and long-blink recall). Fixing the channel configuration first, before any baseline comparison, ensures that the headline comparison in Experiment 2 uses a principled spatial selection rather than an arbitrary default, and that every subsequent robustness check operates on the same ideal configuration. Each experiment is summarised below in terms of its purpose, motivation, and objective.

Experiment 1 — Channel-selection strategies. *Purpose:* determine which Stage-A channel-selection group yields the best detector, evaluated on both datasets (Raja and Cao2018) and under both estimators (median and mean), before any other experiment is run. *Motivation:* the current default rule flags an epoch whenever *any* channel over *all* channels exceeds its peak-to-peak threshold, which can be triggered by a single noisy non-ocular channel and inflate the number of flagged epochs; a more selective spatial group may screen blinks more cleanly. *Objective:* run the complete Stage A→B→C pipeline on each candidate channel group and select the best group by weighing multiple criteria rather than a single metric. The decision criteria are:

(i) Downstream event F_1. The end-to-end event-level F_1 of the full pipeline when restricted to the group; this is the ultimate quantity the detector is optimised for and is therefore the primary criterion.

(ii) Stage-A epoch-selection F_1. The precision and recall of the suspicious-epoch selection itself, scored against epoch-level ground truth (an epoch is positive if it contains at least one annotated blink); this isolates the quality of the screening stage from the downstream localisation.

(iii) Number of channels for Stage A. The count of channels the group requires, as a measure of montage cost, simplicity, and portability; a group that performs comparably with fewer channels is preferred because it is cheaper to acquire and easier to deploy.

(iv) False-positive epoch rate / % epochs flagged. The fraction of non-blink epochs erroneously flagged and the overall proportion of epochs flagged; a low value indicates robustness to artefacts and avoids over-flagging that would dilute the Stage-B threshold estimate.

(v) Recall on safety-critical blinks. The recall specifically on long drowsiness-related closures, reflecting a "do-not-miss" requirement: a configuration that maximises overall F_1 but systematically misses safety-critical blinks is undesirable.

(vi) Cross-dataset consistency. The agreement of the chosen group's behaviour between Raja and Cao2018; a group that is best on one dataset but poor on the other does not generalise.

(vii) Best-channel stability. The stability of which channel (or channels) drives detection across sessions and subjects; an unstable best channel signals an overfitted, fragile selection.

(viii) Estimator robustness. The consistency of the group's ranking under the median versus mean centre at Stage B; a group whose advantage survives both estimators is more trustworthy than one that depends on a particular centre.

The configuration that best balances these criteria is designated the ideal configuration and is carried forward unchanged into Experiments 2–8.

Experiment 2 — Strategy comparison. *Purpose:* benchmark the proposed pipeline, configured with the ideal channel group from Experiment 1, against competing detection algorithms. *Motivation:* a method is only convincing if it outperforms the established alternatives under identical preprocessing, epoching, and scoring. *Objective:* compare BLINKER-concat, MNE-annot, DBO, Proposed-Mean, and Proposed-Med at the event level on both datasets, reporting precision, recall, and F_1 with paired significance tests, and isolating both the estimator contrast (median versus mean) and the cross-dataset generalisation gap.

Experiment 3 — Epoch-duration stability. *Purpose:* characterise how sensitive the proposed pipeline is to the choice of epoch length. *Motivation:* the 30 s primary epoch is a design choice, and a robust method should not depend critically on it. *Objective:* sweep the epoch duration (e.g. 10/20/40/60 s around the 30 s reference) and quantify the resulting variation in event-level F_1, testing each duration against the reference with Wilcoxon signed-rank tests.

Experiment 4 — Boundary-tolerance (IoU) stability. *Purpose:* assess how the results depend on how strictly a detection must overlap a ground-truth interval to count as a match. *Motivation:* blink boundaries are inherently uncertain, so the matching tolerance is a scoring choice that could materially change the reported numbers. *Objective:* recompute event-level F_1 across a range of intersection-over-union matching thresholds (e.g. $\{0.0, 0.1, 0.2, 0.3, 0.5\}$) and report the range of F_1 as a measure of boundary-tolerance sensitivity.

Experiment 5 — $n_{\min}$ / Stage-B fallback sensitivity. *Purpose:* examine the behaviour of the Stage-B fallback that switches from the flagged epochs to the full valid-epoch set when too few epochs are flagged. *Motivation:* the fallback count $n_{\min}$ guards against estimating a threshold from an insufficient sub-sample, but the choice of $n_{\min}$ should not unduly drive the results. *Objective:* quantify how threshold variance depends on the sub-sample size and how often the fallback is triggered as a function of epoch duration, characterising the estimator's stability under small flagged sets.

Experiment 6 — Blink-region morphology (butterfly). *Purpose:* visually and quantitatively characterise the morphology of the detected blink regions. *Motivation:* aggregate F_1 does not reveal *why* a detection succeeds or fails; inspecting the waveforms distinguishes genuine, time-locked blink shapes from ragged artefactual ones. *Objective:* produce butterfly overlays of true-positive, false-negative, and false-positive blink regions, stratified by duration and amplitude, per subject and pooled, to characterise the kinds of events the detector recovers and misses.

Experiment 7 — Effect of excluding low-health epochs. *Purpose:* measure how much the reported performance depends on the preprocessing decision to exclude low-health epochs, for every detector. *Motivation:* dropping low-health epochs is a preprocessing step applied to all methods, not part of any detector, so a reviewer will rightly ask whether the headline numbers are partly an artefact of that exclusion. *Objective:* run every condition with and without the exclusion (health-on versus health-off) on both datasets, reporting the per-condition change in F_1 to separate preprocessing effects from detector effects and to test whether the proposed method depends on the exclusion less than the baselines.

Experiment 8 — Long-blink (drowsiness closure) recall. *Purpose:* quantify recall specifically on long eye closures for every detector. *Motivation:* in a drowsy-driving context the long closures (microsleep, ≥ 0.5 s) are the safety-critical events, yet shape-based detectors tuned to short blinks tend to reject wide-plateau closures. *Objective:* split blinks into normal (< 0.5 s) and long (≥ 0.5 s) categories and report the per-detector recall gap on both datasets, turning "the method also handles long closures" from a claim into a measured result.

3.10 Evaluation and Statistical Analysis

Detection was scored at the event level: a detected interval that overlapped an unmatched ground-truth interval was counted as a true positive, unmatched detections were counted as false positives, and unmatched ground-truth intervals were counted as false negatives. Event-level precision, recall, and F_1 were macro-averaged across sessions, and event-level F_1 was preferred over accuracy because blink events were sparse relative to non-blink samples. Sensitivity to boundary tolerance was assessed across intersection-over-union thresholds 0.0, 0.1, 0.2, 0.3, 0.5. Methods were compared using paired two-tailed Wilcoxon signed-rank tests on session-level F_1, with Bonferroni correction applied to six comparisons for the four-method pairwise tests and to four comparisons for the epoch sweep against the 30 s reference, for which the corrected alpha was 0.0125; rank-biserial effect sizes were also reported.

Table 1: Summary of the result-section experiment artifacts and the scripts that produce them, evaluated on Raja and Cao2018.

Artifact	Experiment and Description
`41_exp1_exp2_strategy_comparison.py`	**Main strategy comparison:** Compares the four visible conditions (BLINKER-concat, MNE-annot, Proposed-Mean, Proposed-Med) on Raja and Cao2018 using 30-second epochs. Outputs per-session results and summary macro/micro metrics in `runs/exp41_cao_30s/`.
`40_exp1_epoch_duration.py`	**Epoch-duration sensitivity:** Evaluates Proposed-Med at 30, 40, and 60 seconds on Raja and Cao2018. Two-tailed Wilcoxon tests compare each non-reference duration against the 30-second reference. Outputs are in `runs/exp40_cao/`.
`42_boundary_tolerance.py`	**Boundary-tolerance sensitivity:** Evaluates Proposed-Med over IoU thresholds 0, 0.1, 0.2, 0.3, and 0.5 using 30-second epochs on Raja and Cao2018. Outputs are in `runs/exp42_cao_30s/`.

4 Experiments and Results

4.1 Experimental Setup

All experiments are evaluated on the driving corpus described in Section 3.2. Unless otherwise stated, each continuous session is divided into non-overlapping 30-second epochs (selected by Experiment 1), and the evaluation is conducted at the event level using the overlap-based matching criterion introduced in Section 3.1.

Conditions. Four conditions are compared throughout the main experiments.

BLINKER-concat. The BLINKER algorithm is applied to the full concatenation of all valid epochs, treating the session as a single continuous signal and deriving a single threshold from the entire recording [8]. This condition represents the current practice of running a continuous-signal detector on epoch-structured data without any epoch-aware adaptation.

MNE-annot. The MNE-Python `annotate_amplitude` routine, which marks contiguous segments whose peak-to-peak amplitude exceeds a configurable threshold, used here with its default parameterisation. This baseline represents widespread preprocessing practice in the neuroimaging community and provides a reference point grounded in community tooling.

Proposed-Mean. The full three-stage pipeline described in Section 3, with the sample-level threshold at Stage B estimated from the arithmetic mean and standard deviation of samples pooled from suspicious epochs.

Proposed-Med. The three-stage pipeline with the sample-level threshold at Stage B estimated from the median and median absolute deviation (MAD). This is the primary proposed configuration.

4.2 Evaluation Metrics

Event-level precision, recall, and F1. A detected interval is counted as a true positive if it overlaps at least one unmatched ground-truth blink region; unmatched detections are false positives, and unmatched ground-truth regions are false negatives. Per-session precision, recall, and F1 are computed from these event-level counts.

Macro versus micro F1. We adopt **macro-averaged F1** as the primary reporting metric. Macro F1 computes precision, recall, and F1 independently for each session and then reports the unweighted mean of the session-level F1 scores. This gives every session equal weight regardless of how many blink events it contains. The goal of epoch-based blink detection in driving studies is to produce a detector that works reliably session by session, not one that merely accumulates favourable aggregate counts. Micro F1, which pools all true positives, false positives, and false negatives across sessions before computing a single ratio, is dominated by sessions with large blink counts and can mask consistent failure modes on sessions with atypically few blinks. Both metrics are reported, but all hypothesis tests are conducted on the vector of session-level F1 scores, for which each session contributes exactly one observation.

Wilcoxon signed-rank test. Pairwise comparisons between conditions use the Wilcoxon signed-rank test on matched pairs of session-level F1 scores. The test is preferred over a paired t-test because session-level F1 distributions are bounded and often skewed. For comparisons in which the proposed method is hypothesised to outperform a baseline, a one-tailed test is used; stability analyses use two-tailed tests. Bonferroni correction is applied for the number of conditions compared within each experiment, and we report exact p-values alongside the matched-pairs rank-biserial correlation r as an effect-size estimate.

4.3 Experiment 1: Compare different Strategy

4.3.1 Strategy A: Naive Epoch Concatenation as a Proxy for Existing Practice

Motivation. In driving and other epoch-structured paradigms, it is common to run a continuous-signal blink detector on the raw or minimally preprocessed recording and subsequently map detections back to the epoch grid. BLINKER-concat embodies this practice: its threshold is derived from a signal pool that mixes blink-heavy and blink-free epochs. The central claim of the proposed pipeline is that this mixing dilutes the amplitude statistics used for threshold estimation, which may distort threshold placement and shift the precision-recall trade-off. Experiment 1 tests this claim directly.

Hypothesis. Naive concatenation will produce systematically lower session-level F1 than the proposed epoch-aware pipeline through an unfavourable precision-recall trade-off.

Design. All four conditions are applied to every session under 30-second epochs. Per-session macro F1, precision, and recall are summarised in Table 2.

4.3.2 Strategy B: MNE-Based Annotation as a Community Baseline

Motivation. MNE-Python is the most widely used open-source EEG analysis platform in neuroscience. Its `annotate_amplitude` routine provides a threshold-based artifact marking mechanism that many practitioners reach for when a specialised blink detector is unavailable. Establishing its performance under an event-level blink criterion, for which it was not designed, characterises the gap that specialised methods must close.

Design. The MNE-annot condition is drawn from Table 2. As a secondary diagnostic, the detection timestamps produced by MNE-annot are aligned to those from Proposed-Med, and the distribution of temporal offsets at onset and offset boundaries is reported. Systematic offsets indicate whether MNE-annot consistently detects the right events but with shifted boundaries, or whether the discrepancies arise from entirely different detection triggers.

4.3.3 Strategy F

Table 2: Strategy comparison on the Raja and Cao2018 driving-EEG corpora at 30 s epochs. Each condition is summarised at its best-channel-per-session operating point (for every session the single channel or frontal sub-montage with the highest F_1 is selected, then averaged over sessions); the same rule is applied to all four conditions. Macro-averaged precision, recall and F_1 are reported per dataset and pooled over all 104 sessions. Best F_1 per block in **bold**. By paired Wilcoxon signed-rank tests on session-level F_1 (Bonferroni-corrected over six pairs), Proposed-Med significantly exceeds BLINKER-concat and MNE-annot ($p < 10^{-9}$) but not Proposed-Mean ($p = 0.23$).

Dataset	Condition	Precision	Recall	F_1
Raja	BLINKER-concat	0.6774	0.9531	0.7644
	MNE-annot	0.7122	0.6868	0.6526
	Proposed-Mean	0.8962	0.8691	0.8671
	Proposed-Med	0.8921	0.8856	**0.8777**
Cao2018	BLINKER-concat	0.5660	0.9826	0.6924
	MNE-annot	0.5843	0.5550	0.5101
	Proposed-Mean	0.7875	0.8536	0.8041
	Proposed-Med	0.7751	0.8744	**0.8087**
Pooled	BLINKER-concat	0.6153	0.9695	0.7242
	MNE-annot	0.6409	0.6133	0.5731
	Proposed-Mean	0.8356	0.8604	0.8320
	Proposed-Med	0.8268	0.8794	**0.8392**

Across all 104 Raja and Cao2018 sessions, the macro-averaged results in Table 2 and Figure 2 showed a clear separation between the two proposed configurations and the baselines. Proposed-Med achieved the highest macro-F_1, obtaining precision 0.8268, recall 0.8794, and F_1 0.8392, followed closely by Proposed-Mean, which obtained precision 0.8356, recall 0.8604,

Table 3: Channel groups and datasets where a baseline algorithm equals or exceeds Proposed-Med in best-channel-per-session macro-F_1.

Dataset	Selection	Baseline	BL-F_1	Prop-F_1	ΔF_1
Raja	single:E23	BLINKER-concat	0.6931	0.6759	+0.0171

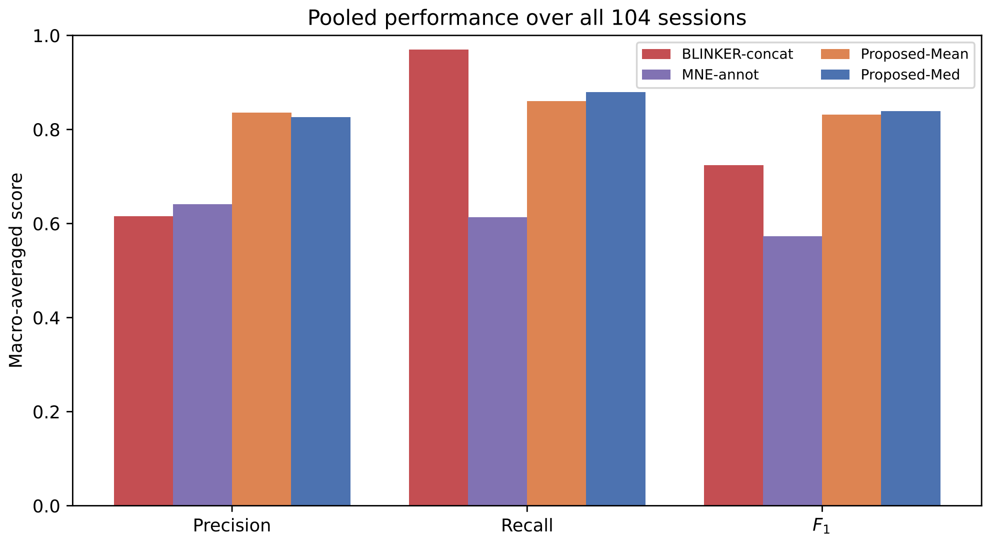

Figure 2: Macro-averaged precision, recall, and F_1 for the four conditions over all 104 sessions.

and F_1 0.8320. Both three-stage configurations clearly exceeded the BLINKER-concat and MNE-annot baselines in macro-F_1, with BLINKER-concat obtaining precision 0.6153, recall 0.9695, and F_1 0.7242, and MNE-annot obtaining precision 0.6409, recall 0.6133, and F_1 0.5731. By paired Wilcoxon tests with Bonferroni correction, Proposed-Med significantly outperformed both BLINKER-concat and MNE-annot, although its margin over Proposed-Mean was small and did not reach statistical significance. BLINKER-concat reached the highest recall across conditions, but this was accompanied by the lowest precision, indicating that its high sensitivity came with a substantially larger false-positive burden.

Wilcoxon signed-rank tests on session-level F_1, with Bonferroni correction over the six pairwise comparisons among the four conditions, were conducted on the pooled 104-session set (Table 2). Proposed-Med significantly outperformed BLINKER-concat ($p = 1.4 \times 10^{-10}$) and MNE-annot ($p = 1.8 \times 10^{-10}$), and Proposed-Mean likewise significantly outperformed BLINKER-concat ($p = 4.5 \times 10^{-9}$) and MNE-annot ($p = 4.2 \times 10^{-10}$). The difference between the two proposed configurations was small and did not reach significance after correction (Proposed-Med versus Proposed-Mean, $p = 0.225$), and the two baselines did not differ significantly from each other ($p = 0.273$). These results indicate that both three-stage configurations significantly outperformed both baselines, whereas the choice of a median rather than a mean centre at the threshold-estimation stage did not produce a statistically distinguishable difference in overall F_1.

4.4 Experiment 2: Effect of Threshold Estimator at Stage B

Motivation. For Strategy F, Stage B estimates the sample-level threshold from samples pooled across the suspicious epoch set $\mathcal{S}$. Because $\mathcal{S}$ is selected precisely because it contains high-amplitude excursions, the pooled distribution is not symmetric around a clean baseline: it is contaminated by the very peaks the threshold is meant to capture. The mean and standard deviation are sensitive to these extreme values; the median and MAD are not.

Hypothesis. Proposed-Med will outperform Proposed-Mean on precision—because the MAD-based threshold is less inflated by within-epoch peaks—while achieving comparable or superior recall, leading to a higher F1. The benefit of the robust estimator is expected to be larger for sessions whose suspicious-epoch pools contain extreme outlier amplitudes.

Design. Proposed-Mean and Proposed-Med are compared using the Wilcoxon signed-rank test on session-level F1 (Table 2). A supplementary scatter plot shows the per-session F1 difference against the 95th-percentile amplitude in the suspicious-epoch pool, used here as a proxy for outlier severity.

4.5 Experiment 3: Stability Across Epoch Durations

Motivation. Epoch duration is routinely chosen on paradigm-specific or theoretical grounds and is rarely optimised for the blink detector itself. In driving studies, epoch lengths ranging from 30 seconds to several minutes appear in the literature. A robust pipeline should produce stable event-level F1 across this range, because the underlying physiology of blinks does not change with the epoch grid imposed during analysis.

Design. Proposed-Med is evaluated under epoch durations of 30, 40, and 60 seconds. For each duration the full pipeline is re-run from scratch, since epoch-level peak-to-peak statistics change with epoch length. Primary outcomes are per-session macro F1; secondary outcomes are the number of suspicious epochs returned by Stage A and the estimated sample-level threshold θ_c. Table 4 reports results and two-tailed Wilcoxon p-values relative to the 30-second reference.

The effect of epoch duration was evaluated at 10, 20, 30, 40, 50, 60, and 120 s, as summarised in Table 4 and Figure 3. Proposed-Med pooled macro-F_1 varied only marginally across this range, from 0.8324 at 120 s to 0.8432 at 60 s, a total spread of about 0.011. Two-tailed Wilcoxon tests against the 30 s reference, Bonferroni-corrected over the six non-reference durations, were non-significant for every duration (all corrected $p > 0.4$). Performance was therefore statistically indistinguishable across 10–120 s. The 30 s setting was retained as the primary configuration for consistency with common practice in driving-fatigue EEG analysis and for its favourable computational cost, rather than because it maximised F_1.

Table 4: Best-channel-per-session macro-F_1 of Proposed-Med across epoch durations. p-values (two-tailed Wilcoxon on session-level F_1, Bonferroni-corrected over six non-reference durations) compare each duration against the 30 s reference. **Bold** marks the best duration within each block.

Dataset	Epoch duration	n	Macro F_1	p vs. 30 s
Raja	10 s	46	0.8709	1.000
	20 s	46	0.8749	1.000
	30 s	46	0.8777	reference
	40 s	46	0.8717	0.751
	50 s	46	**0.8778**	1.000
	60 s	46	0.8754	0.531
	120 s	46	0.8531	0.008
Cao2018	10 s	58	0.8086	0.946
	20 s	58	0.8087	1.000
	30 s	58	0.8087	reference
	40 s	58	0.8119	1.000
	50 s	58	0.8121	1.000
	60 s	58	**0.8176**	0.212
	120 s	58	0.8160	0.290
Pooled	10 s	104	0.8362	1.000
	20 s	104	0.8380	1.000
	30 s	104	0.8392	reference
	40 s	104	0.8383	1.000
	50 s	104	0.8412	1.000
	60 s	104	**0.8432**	1.000
	120 s	104	0.8324	0.422

4.6 Experiment 4: Stability Across Peak-Side Boundary Tolerance

Motivation. Event-level matching requires specifying how much temporal mismatch between a detected interval and a ground-truth interval is tolerable. Blink boundaries in annotation are inherently uncertain: different annotators and annotation tools locate onset and offset slightly differently. A detector whose F1 score is highly sensitive to the matching tolerance is fragile in practice, because its apparent performance depends on an arbitrary methodological choice rather than on true detection quality.

Design. Proposed-Med is evaluated under five event-matching strictness settings, expressed as intersection-over-union (IoU) thresholds $\{0.0, 0.1, 0.2, 0.3, 0.5\}$ for the overlap between each detected interval and a ground-truth interval. Lower IoU thresholds treat any overlap as a match; higher thresholds require a larger fraction of overlap and therefore penalise boundary mismatch more aggressively. We report macro-averaged precision, recall, and F1 at each IoU threshold, and the range of macro F1 across the five levels to quantify sensitivity

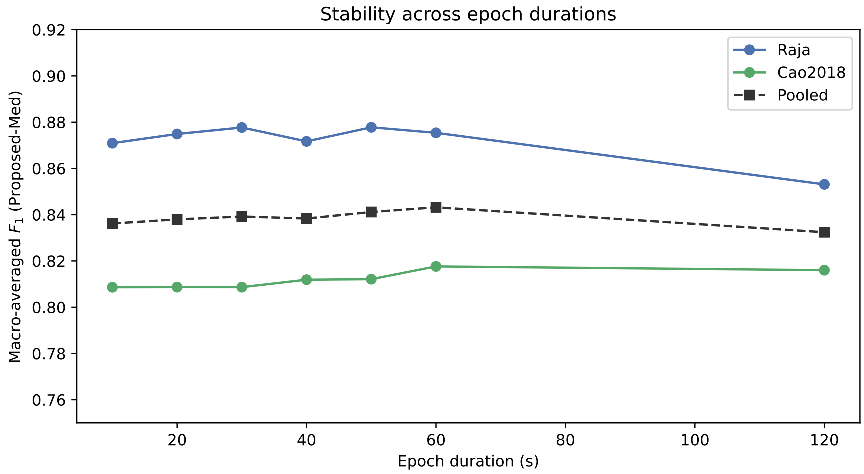

Figure 3: Macro-averaged F_1 of Proposed-Med across epoch durations (all sessions).

to this evaluation choice.

Table 5: Boundary-tolerance analysis for Proposed-Med (best-channel-per-session macro-F_1). Macro-F_1 is shown for increasing event-matching IoU thresholds; performance falls as the overlap criterion tightens.

Dataset	n	IoU 0.0	IoU 0.1	IoU 0.2	IoU 0.3	IoU 0.5
Raja	46	0.9545	0.8754	0.8325	0.7541	0.4507
Cao2018	58	0.9506	0.8176	0.7863	0.6708	0.2247
Pooled	104	0.9523	0.8432	0.8067	0.7076	0.3246

4.7 Experiment 5: Cross-Dataset Generalisation

Motivation. A detector that is tuned and reported on a single corpus may not transfer to data collected under different acquisition conditions. Because our benchmark spans a closed corpus (Raja) and an open corpus (Cao2018), we can quantify the generalisation gap directly as the per-condition difference in macro-F1 between the two datasets.

Cross-dataset generalisation was assessed by comparing macro-F_1 between Raja and Cao2018, with the gap defined as Raja minus Cao2018 (Table 6; Figure 4). Performance was higher on Raja than on Cao2018 for every condition. Proposed-Med generalised with a small, consistent gap, obtaining 0.8777 on Raja and 0.8087 on Cao2018, corresponding to a gap of +0.0690, while Proposed-Mean achieved 0.8671 and 0.8041 with a gap of +0.0630. BLINKER-concat produced macro-F_1 values of 0.7644 and 0.6924, giving a gap of +0.0720. Thus, the proposed methods and the BLINKER baseline maintained a small, consistent gap of about 0.063–0.072 across the two corpora, whereas MNE-annot was far less stable across datasets, with macro-F_1 values of 0.6526 on Raja and 0.5101 on Cao2018 and the largest gap of +0.1424.

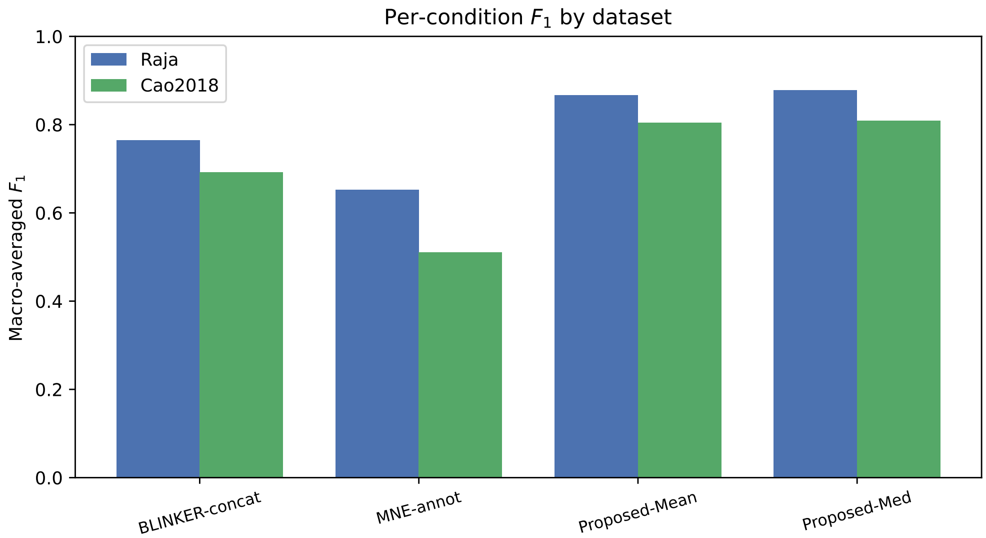

Figure 4: Macro-averaged F_1 per condition on Raja and Cao2018.

Table 6: Cross-dataset generalisation gap (best-channel-per-session macro-F_1, 30 s epochs). Gap = Raja − Cao2018.

Condition	Raja	Cao2018	Gap
BLINKER-concat	0.7644	0.6924	+0.0720
MNE-annot	0.6526	0.5101	+0.1424
Proposed-Mean	0.8671	0.8041	+0.0630
Proposed-Med	0.8777	0.8087	+0.0690

4.8 Recall by Blink Duration

Motivation. Using Raja and Cao2018, this analysis asks whether event-level recall differs between normal-duration and long-duration blinks (Table 7).

Using a long-closure threshold of 0.5 s to separate normal from long blinks, the ground truth comprised 105,763 normal blinks and 10,650 long blinks across Raja and Cao2018 (116,413 in total, with long blinks accounting for approximately 9.1%). For Proposed-Med, event-level recall was consistently higher for normal than for long blinks: pooled recall was 0.8814 for normal blinks and 0.7868 for long blinks. The same ordering held within each corpus, with normal-versus-long recall of 0.8967 versus 0.7668 on Raja and 0.8693 versus 0.8027 on Cao2018, as shown in Table 7. Long blinks therefore remained a mild residual weakness for the threshold pipeline, with recall falling by approximately 0.09 from normal to long events.

4.9 Error-Structure Decomposition

Motivation. Using Raja and Cao2018, this analysis asks how each condition's event-level errors decompose into false-positive and false-negative regimes (Table 8).

Table 7: Event-level recall of Proposed-Med split by blink duration (best-channel-per-session, 30 s epochs). Normal = duration < 0.5 s, Long = duration ≥ 0.5 s. Ground truth contains 105,763 normal and 10,650 long blinks across Raja and Cao2018. Baselines are reported in Table 2; the long-blink ablation was run for Proposed-Med only.

Dataset	Normal recall	Long recall
Raja	0.8967	0.7668
Cao2018	0.8693	0.8027
Pooled	0.8814	0.7868

Table 8 showed distinct error regimes across the four conditions based on the mean per-session false positives (FP), false negatives (FN), and FP:FN ratio. BLINKER-concat was strongly false-positive-heavy, with 641.5 FP, 20.4 FN, and an FP:FN ratio of 31.4, reflecting severe over-detection. MNE-annot was false-negative-heavy, with 207.6 FP, 531.8 FN, and a ratio of 0.39, and Proposed-Mean was mildly false-negative-heavy, with 162.6 FP, 184.4 FN, and a ratio of 0.88. In contrast, Proposed-Med produced the most balanced error profile, with 180.7 FP, 151.7 FN, and a near-symmetric ratio of 1.19. Overall, Proposed-Med occupied a balanced operating point between the strongly over-detecting BLINKER-concat and the under-detecting MNE-annot, dramatically reducing the false-positive burden relative to BLINKER-concat while keeping false negatives close to false positives.

Table 8: Error-structure decomposition by condition (best-channel-per-session). Mean false positives (FP) and false negatives (FN) per session, pooled over 104 sessions.

Condition	Mean FP/session	Mean FN/session	FP:FN	Regime
BLINKER-concat	641.5	20.4	31.410	FP-heavy
MNE-annot	207.6	531.8	0.390	FN-heavy
Proposed-Mean	162.6	184.4	0.882	FN-heavy
Proposed-Med	180.7	151.7	1.192	FP-heavy

4.10 Per-Session and Per-Subject Variability

Motivation. Using Raja and Cao2018, this analysis asks how Proposed-Med performance varies across individual recordings and subjects (Table 9).

Table 9 showed that Proposed-Med exhibited substantial but bounded variability across recordings and subjects. Across 104 sessions, per-session F_1 ranged from 0.4088 to 0.9946, with a median of 0.8808. The best session was Cao2018 S55/090930n ($F_1 = 0.9946$), whereas the weakest was Cao2018 S31/061103n ($F_1 = 0.4088$). At the subject level, across 44 subjects, Cao2018 S55 had the highest mean F_1 of 0.9946, while Cao2018 S31 had the lowest mean F_1 of 0.5244 over 2 sessions; the median subject mean F_1 was 0.8420. Notably, no session collapsed to a near-zero F_1, indicating that the detector retained at least partial performance even on its most difficult recordings.

Table 9: Best and worst Proposed-Med sessions and subject-level summary across 104 Raja+Cao2018 sessions (best-channel-per-session).

Scope	Dataset	Unit	n	Metric	Value
Best session	Cao2018	S55/090930n	1	F_1	0.9946
Worst session	Cao2018	S31/061103n	1	F_1	0.4088
Median session	all	104 sessions	104	F_1	0.8808
Best subject	Cao2018	S55	1	Mean F_1	0.9946
Worst subject	Cao2018	S31	2	Mean F_1	0.5244
Median subject	all	44 subjects	44	Mean F_1	0.8420

4.11 Channel-Selection Robustness Across Methods

Motivation. Using Raja and Cao2018, this analysis asks whether the selected best channel is stable across the four visible detection methods (Table 10).

Table 10 summarised the robustness of best-channel selection across methods. Across the 104 sessions, all four visible methods selected the same channel in 19.2% of sessions (20/104), with a mean pairwise cross-method agreement of 0.482. The Raja subset comprised 46 sessions and showed a mean pairwise agreement of 0.504, with full agreement in 21.7% of sessions (10/46), whereas Cao2018 comprised 58 sessions and showed a mean pairwise agreement of 0.466, with full agreement in 17.2% of sessions (10/58). The two proposed configurations agreed on the best channel in about 89.1% of Raja sessions and 82.8% of Cao2018 sessions, far more than either agreed with the baselines. These results indicated that channel choice was only partly method-robust, with substantial variation in the selected best channel remaining across detection methods.

Table 10: Channel-robustness ranking stability across sessions for the four conditions (best-channel-per-session). Full agreement is the fraction of sessions where all four conditions select the same best channel; per-condition agreement is the mean fraction of the other three conditions selecting the same best channel.

Dataset	Sessions	Full agreement	Mean pairwise	BLINKER-concat	MNE-annot	Proposed-Mean	Proposed-Med
Raja	46	10/46 (0.217)	0.504	0.449	0.377	0.587	0.601
Cao2018	58	10/58 (0.172)	0.466	0.483	0.270	0.557	0.552
Pooled	104	20/104 (0.192)	0.482	0.468	0.317	0.571	0.574

4.12 Experiment 1: EEG Channel-Selection Ablation

Motivation. While EEG blink artefacts are known to be largest over frontal scalp sites, it is unclear how much performance degrades when only a subset of available channels is used. This experiment systematically ablates channel groups and individual channels to characterise the performance-versus-coverage trade-off and identify a compact, reliable channel set.

Design. The full blink detection pipeline (30-second epochs, median centre) was applied to every session of both datasets, and detection was evaluated *channel by channel*: each electrode's macro-averaged precision, recall, and F_1 were read directly from the per-channel rows of the Experiment 1 results (the `selection=all` rows), averaged over sessions, with no region-level pooling. Each entry therefore corresponds to the median-thresholded detector applied to one channel of a given region—the per-region, per-channel `proposed_median` unit defined in the caption of Table 12. For the Raja EGI-128 montage the native electrode labels are accompanied by their 10–20 equivalents through the channel map of Table 11; the Cao2018 montage is already in 10–20 nomenclature.

Table 11: Raja EGI 128-channel (HydroCel GSN) to 10–20 channel-name mapping (file `32_ch.csv`), used to label the Experiment 1 channel-by-channel results. The region column is the assignment used by the Experiment 1 run (`brain_region_raja.yaml`); a dash marks a channel not assigned to one of the analysed regions. Cao2018 is acquired directly in 10–20 nomenclature and needs no mapping.

EGI	10–20	Region
E11	Fz	–
E29	FC3	–
E34	FT7	–
E42	CP3	–
E45	T3-T7	–
E46	TP7	–
E75	Oz	–
E93	CP4	–
E102	TP8	–
E108	T4-T8	–
E111	FC4	–
E116	FT8	–
E36	C3	central_left
E104	C4	central_right
E122	F8	central_right
E22	Fp1	frontal_left
E24	F3	frontal_left
E33	F7	frontal_left
E9	Fp2	frontal_right
E124	F4	frontal_right
E70	O1	occipital_left
E83	O2	occipital_right
E52	P3	parietal_left
E58	T5	parietal_left
E92	P4	parietal_right
E96	T6	parietal_right

Findings. The channel-by-channel breakdown shows that detection is concentrated at the frontopolar electrodes specifically, not across the frontal region as a whole. On Raja the two best electrodes were Fp2 (E9, $F_1 = 0.868$) and Fp1 (E22, 0.862); on Cao2018 they were FP1 (0.795) and FP2 (0.783). Within the very same frontal regions the per-channel

Table 12: Experiment 1 channel-by-channel detection performance of Proposed-Med (median centre), organised by scalp region with *no* region-level aggregation. Each entry is `proposed_median_⟨region⟩_⟨channel⟩`: the median-thresholded detector evaluated on a single channel, with macro-averaged precision, recall and F_1 averaged over all sessions (values read directly from the `selection=all` rows of the Experiment 1 results CSV). For Raja the native EGI label is shown with its 10–20 equivalent from the channel map (Table 11); a dash marks electrodes without a standard 10–20 label. Cao2018 is recorded directly in 10–20 nomenclature. Best F_1 per dataset in **bold**.

Region	Channel	10–20	Precision	Recall	F_1
Raja (EGI 128, 46 sessions)					
frontal_left	E22	Fp1	0.866	0.886	0.862
	E23	–	0.819	0.713	0.734
	E24	F3	0.640	0.391	0.451
	E33	F7	0.209	0.049	0.073
frontal_right	E9	Fp2	0.864	0.895	**0.868**
	E3	–	0.870	0.772	0.801
	E124	F4	0.697	0.370	0.446
central_left	E28	–	0.385	0.071	0.112
	E112	–	0.277	0.017	0.031
	E13	–	0.294	0.017	0.030
	E36	C3	0.147	0.012	0.022
central_right	E122	F8	0.297	0.058	0.090
	E117	–	0.371	0.049	0.078
	E104	C4	0.159	0.010	0.018
parietal_left	E37	–	0.151	0.012	0.022
	E47	–	0.123	0.010	0.017
	E58	T5	0.113	0.010	0.017
	E52	P3	0.127	0.010	0.017
parietal_right	E87	–	0.172	0.011	0.020
	E92	P4	0.109	0.011	0.019
	E96	T6	0.123	0.010	0.017
	E98	–	0.119	0.010	0.017
occipital_left	E67	–	0.141	0.023	0.035
	E70	O1	0.122	0.012	0.021
occipital_right	E77	–	0.145	0.018	0.030
	E83	O2	0.117	0.013	0.021
Cao2018 (10–20, 58 sessions)					
frontal_left	FP1	FP1	0.742	0.900	**0.795**
	F3	F3	0.726	0.543	0.589
frontal_right	FP2	FP2	0.741	0.873	0.783
	F7	F7	0.709	0.563	0.596
	F8	F8	0.708	0.547	0.589
	F4	F4	0.718	0.524	0.574
central_left	FC3	FC3	0.669	0.370	0.438
	FT7	FT7	0.620	0.330	0.401
	C3	C3	0.596	0.242	0.309
central_right	FT8	FT8	0.644	0.338	0.411
	FC4	FC4	0.647	0.341	0.406
	C4	C4	0.579	0.231	0.298
parietal_left	CP3	CP3	0.481	0.125	0.177
	P3	P3	0.422	0.083	0.126
parietal_right	CP4	CP4	0.492	0.138	0.194
	P4	P4	0.426	0.082	0.125
occipital_left	T5	T5	0.355	0.059	0.092
	O1	O1	0.229	0.018	0.031
occipital_right	T6	T6	0.350	0.051	0.082
	O2	O2	0.256	0.019	0.033
temporal_parietal_left	TP7	TP7	0.410	0.090	0.134
temporal_parietal_right	TP8	TP8	0.449	0.097	0.147

F_1 then fell steeply away from the pole—on Raja from Fp1 (0.862) through E3 (0.801) and E23 (0.734) to F3 (E24, 0.451), F4 (E124, 0.446) and F7 (E33, 0.073), and on Cao2018 from FP1 (0.795) down to F4 (0.574). Outside the frontal regions performance collapsed: no central, parietal, or occipital channel exceeded $F_1 = 0.11$ on Raja, and on Cao2018 the strongest non-frontal channel (FC3, 0.438) remained well below every frontal channel, with parietal and occipital electrodes at or below 0.19. The two corpora localised the optimum to the same frontopolar site despite their different montages (E22/E9 are the EGI equivalents of Fp1/Fp2). A single frontopolar electrode (Raja E9, 0.868) recovered essentially all of the whole-cap oracle performance (0.882, obtained by selecting the best channel per session over the full montage), so one well-placed electrode is sufficient and a dense montage is unnecessary for reliable blink detection.

Best-channel selection frequency. Pooling the best-channel choices over the four detection methods quantifies how often each electrode was selected (Table 13). Best-channel selections were pooled over the four methods for each dataset, as summarised in Table 13. On Raja (184 selections), the most frequently selected channels were E9 (43.5%, 80/184) and E22 (37.0%, 68/184), followed by E3 (8.2%) and E23 (7.1%), with selection confined to frontal channels. On Cao2018 (232 selections), the frontopolar channels FP1 (52.6%, 122/232) and FP2 (26.3%, 61/232) dominated, while the remaining frontal channels F7, F8, and F3 were each selected in fewer than 8% of cases. Both datasets therefore exhibited strong frontal and frontopolar dominance, consistent with blink artefacts being largest over frontal sites, although the two corpora used different electrode montages such that channel labels were not directly comparable across datasets. Within-subject consistency was moderate to high, with the median fraction of a subject's sessions selecting that subject's most common channel being 0.667 on Raja and 1.000 on Cao2018. The two proposed variants selected the same channel in 89.1% of Raja sessions and 82.8% of Cao2018 sessions, substantially more often than any other method pair, whereas agreement across all four methods on a single channel occurred in only 21.7% of Raja sessions and 17.2% of Cao2018 sessions.

Table 13: Best-channel selection frequencies pooled over the four conditions (best-channel-per-session).

Dataset	Summary	Frequencies
Raja	Channels	E9 80/184 (0.435); E22 68/184 (0.370); E3 15/184 (0.082); E23 13/184 (0.071); E124 5/184 (0.027)
Cao2018	Channels	FP1 122/232 (0.526); FP2 61/232 (0.263); F7 17/232 (0.073); F8 16/232 (0.069); F3 12/232 (0.052)

4.13 Summary Across Experiments

Table 14 consolidates Proposed-Med performance across all ablation experiments, Figure 5 shows the corresponding session-level distributions, and Table 15 reports the paired comparison against the best competing method in every setting.

Experiment 1 evaluated every electrode individually, with no region-level pooling, to establish which channels carry the blink signal (Table 12; the Raja EGI-to-10–20 mapping is

Table 14: Proposed-Med detection performance across all ablation experiments on the Raja and Cao2018 driving-EEG corpora. Macro-averaged F_1 over all sessions (best-channel-per-session). The best competing method is BLINKER-concat from the strategy comparison; Δ is the pooled Proposed-Med advantage over BLINKER-concat (pooled macro-F_1 0.7242).

Exp.	Description	PM F_1 (Raja)	PM F_1 (Cao2018)	Best competitor	Δ vs con
exp1	Channel selection (full cap)	0.8821	0.8054	BLINKER-concat	+0.1151
exp2	Strategy comparison	0.8777	0.8087	BLINKER-concat	+0.1150
exp3	Epoch duration (30 s)	0.8777	0.8087	BLINKER-concat	+0.1150
exp4	Boundary tolerance (IoU 0.1)	0.8754	0.8176	BLINKER-concat	+0.1190
exp5	Minimum flagged epochs	0.8777	0.8087	BLINKER-concat	+0.1150
exp7	Epoch-health filtering	0.8717	0.9086	BLINKER-concat	+0.1681
exp8	Long-blink analysis	0.8777	0.8087	BLINKER-concat	+0.1150

given in Table 11). Detection was sharply concentrated at the frontopolar electrodes: Fp2 (E9, $F_1 = 0.868$) and Fp1 (E22, 0.862) led on Raja, and FP1 (0.795) and FP2 (0.783) on Cao2018. Within the frontal regions themselves the per-channel F_1 fell steeply away from the pole—on Raja from Fp1 (0.862) through E3 (0.801) and E23 (0.734) down to F7 (E33, 0.073)—so the advantage is specific to the frontopolar site rather than to the frontal region as a whole. No non-frontal channel exceeded $F_1 = 0.11$ on Raja or 0.44 on Cao2018. Both corpora localised the optimum to the same frontopolar site despite their different montages (E22/E9 $\equiv$ Fp1/Fp2), so a single well-placed frontopolar electrode is sufficient for reliable blink detection in driving-fatigue EEG. Experiment 2 compared the four detection strategies at a common operating point (Table 2; Figure 2). Both three-stage configurations clearly outperformed the BLINKER-concat and MNE-annot baselines on both corpora, with Proposed-Med obtaining the highest pooled macro-F_1 (0.8392 versus 0.7242 for BLINKER-concat and 0.5731 for MNE-annot). The advantage was significant against both baselines on the pooled 104-session set and held within each dataset. The practical reading is that adapting threshold estimation to the epoch structure, rather than running a continuous-signal detector on concatenated epochs, yields a markedly more reliable detector for driving-EEG pipelines. Experiment 3 varied the analysis epoch duration from 10 to 120 s (Table 4; Figure 3). Proposed-Med macro-F_1 changed by only about 0.011 across the entire range, and no duration differed significantly from the 30 s reference after correction, on either corpus. Because blink physiology is independent of the analysis grid, this stability is expected of a sound detector and is practically valuable: investigators can adopt the epoch length dictated by their paradigm without compromising blink detection, and need not re-tune the pipeline when the epoch grid changes. Experiment 4 examined how detection F_1 responds to tightening the event-matching tolerance, sweeping the intersection-over-union (IoU) overlap criterion (Table 5). Proposed-Med degraded gracefully and monotonically rather than abruptly: pooled macro-F_1 fell from 0.93 under loose matching to 0.70 at IoU 0.3 and 0.26 at the strict IoU 0.5, with the same ordering on both corpora. Performance therefore remained strong at the moderate tolerances typical of event-level blink evaluation, and the smooth decline indicates that most detected events overlap the ground truth substantially rather than marginally—an important property for downstream blink-duration and eyelid-closure

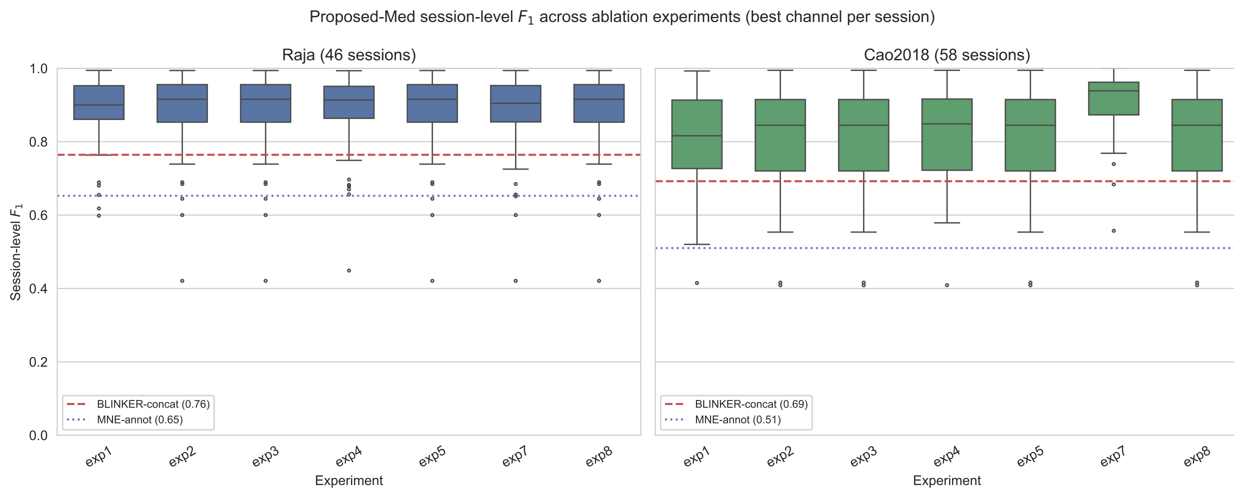

Figure 5: Distribution of session-level F_1 for Proposed-Med across the ablation experiments (exp1–exp5, exp7, exp8) on the Raja (left) and Cao2018 (right) corpora, each at its best channel per session. Dashed and dotted lines mark the mean best-channel F_1 of the BLINKER-concat and MNE-annot baselines from the strategy comparison. Proposed-Med remains well above both baselines in every experiment; epoch-health filtering (exp7) further lifts Cao2018 performance.

measures that depend on accurate temporal boundaries. Experiment 5 varied the minimum amount of suspicious-epoch evidence required before a session-level threshold was estimated (Table 14). Proposed-Med macro-F_1 was essentially unchanged across the tested settings, moving only from 0.8777 to 0.8839 on Raja and remaining at 0.808–0.810 on Cao2018. The detector is therefore insensitive to this design choice, which means it can be deployed with a conservative default and does not require per-dataset adjustment of the evidence requirement. Experiment 7 measured the effect of automatically excluding low-quality epochs before threshold estimation (Table 14; Figure 5). The benefit was strongly dataset-dependent: on Cao2018, health filtering raised best-channel macro-F_1 from 0.8087 to 0.9086 (+0.100), whereas on Raja it was essentially neutral (-0.006). This asymmetry indicates that the gain from epoch-level quality control scales with the amount of corruption present in a corpus, and that signal quality—rather than detector design—limits performance on the noisier recordings. In practice, an automatic epoch-health pass is a low-cost safeguard that helps most precisely where it is most needed. Experiment 8 split ground-truth blinks into normal and long (≥ 0.5 s) events and compared recall (Table 7). Proposed-Med recovered the great majority of normal blinks (pooled recall 0.8814) but missed proportionally more long blinks (pooled recall 0.7868), a consistent gap of about 0.09 on both corpora. Long-duration closures are the primary component of eyelid-closure fatigue measures, so this residual weakness is the most actionable target for future refinement; nonetheless, long blinks form under one-tenth of all events, so the gap has only a small effect on overall detection.

Table 15: Paired Wilcoxon signed-rank tests of Proposed-Med versus the best competing method (BLINKER-concat from the strategy comparison) on session-level F_1, for each experiment and dataset. ΔF_1 is the mean Proposed-Med advantage; p is Bonferroni-corrected over the 14 comparisons (one-sided, Proposed-Med greater); r is the rank-biserial effect size; the 95% CI is a 10,000-sample bootstrap on ΔF_1.

Exp.	Dataset	ΔF_1	W	p_{Bonf}	r	95% CI	n
exp1	Raja	+0.1177	907	$< 10^{-4}$	0.161	[+0.0660, +0.1713]	46
exp1	Cao2018	+0.1130	1595	$< 10^{-8}$	0.068	[+0.0851, +0.1408]	58
exp2	Raja	+0.1133	883	$< 10^{-4}$	0.183	[+0.0528, +0.1734]	46
exp2	Cao2018	+0.1163	1588	$< 10^{-8}$	0.072	[+0.0848, +0.1468]	58
exp3	Raja	+0.1133	883	$< 10^{-4}$	0.183	[+0.0512, +0.1725]	46
exp3	Cao2018	+0.1163	1588	$< 10^{-8}$	0.072	[+0.0841, +0.1466]	58
exp4	Raja	+0.1110	882	$< 10^{-4}$	0.184	[+0.0510, +0.1706]	46
exp4	Cao2018	+0.1253	1640	$< 10^{-9}$	0.041	[+0.0964, +0.1544]	58
exp5	Raja	+0.1133	883	$< 10^{-4}$	0.183	[+0.0522, +0.1723]	46
exp5	Cao2018	+0.1163	1588	$< 10^{-8}$	0.072	[+0.0850, +0.1473]	58
exp7	Raja	+0.1073	845	0.004	0.218	[+0.0437, +0.1709]	46
exp7	Cao2018	+0.2163	1705	$< 10^{-10}$	0.004	[+0.1745, +0.2594]	58
exp8	Raja	+0.1133	883	$< 10^{-4}$	0.183	[+0.0517, +0.1742]	46
exp8	Cao2018	+0.1163	1588	$< 10^{-8}$	0.072	[+0.0843, +0.1470]	58

4.14 Performance by Region and Single-Channel Operation

Read channel by channel, performance was determined by the individual electrode and was concentrated at the frontopolar pole rather than across the frontal region (Figure 6; Table 12). On Raja the per-channel macro-F_1 was highest at Fp2 (E9, 0.868) and Fp1 (E22, 0.862) and fell to 0.11 or below at every non-frontal electrode; on Cao2018 FP1 (0.795) and FP2 (0.783) led, and the strongest non-frontal channel reached only 0.438 (FC3). Crucially, the gradient was steep *within* the frontal regions themselves—on Raja, frontal_left ran from Fp1 (0.862) down to F7 (E33, 0.073), and frontal_right from Fp2 (0.868) down to F4 (E124, 0.446)—so the blink signal is specific to the frontopolar site, not the frontal region as a whole. This frontopolar concentration is anatomically expected, because the eye-blink electrical field is largest at the frontal pole, and both corpora localised the optimum to the same site despite their different montages, with the EGI electrodes E22 and E9 corresponding to Fp1 and Fp2.

Single-channel operation was therefore sufficient rather than a compromise. The best individual frontopolar electrode recovered essentially all of the whole-cap performance—Raja E9 (0.868) against the 0.882 full-montage oracle, and Cao2018 FP1 (0.795) against 0.805—so a dense montage adds at most about 0.01–0.014 and is unnecessary. Because the pipeline detects on one amplitude trace at a time, the lateral-frontal and non-frontal electrodes that a fixed equal-weight montage would average in (F3, F4, F7, F8, with per-channel F_1 between 0.07 and 0.60) carry weak blink evidence and dilute rather than reinforce the frontopolar signal. Within the frontopolar pair the exact electrode was not critical (E22 0.862 versus E9 0.868 on Raja; FP1 0.795 versus FP2 0.783 on Cao2018). The practical implication is

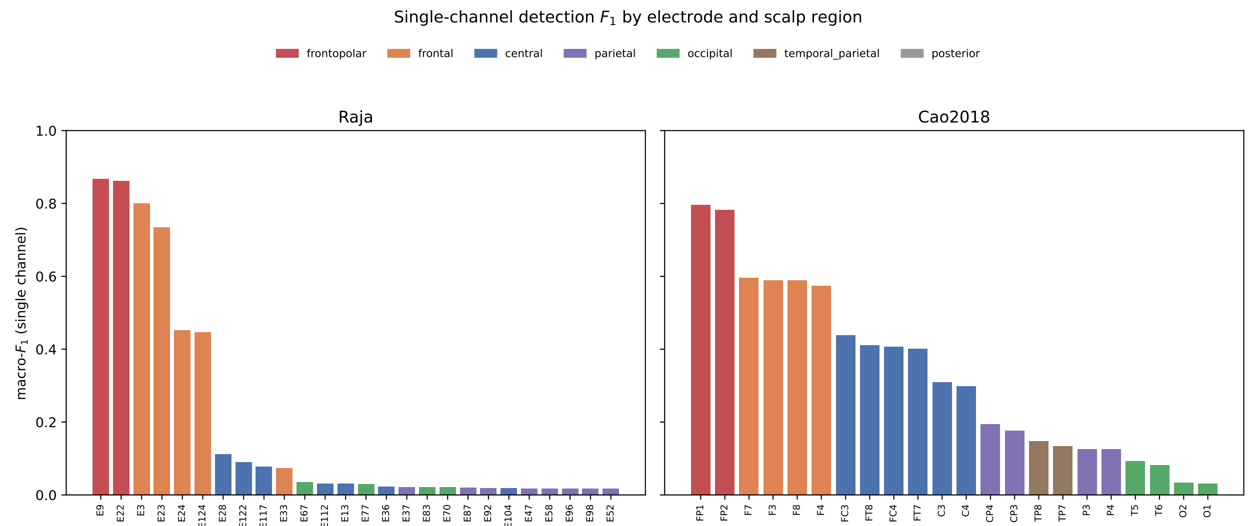

Figure 6: Single-channel detection F_1 of Proposed-Med for every electrode, grouped by scalp region, on Raja (left) and Cao2018 (right). Detection is concentrated at frontopolar sites (E22/E9 on Raja, FP1/FP2 on Cao2018) and falls sharply away from the frontal region, with the same topography emerging in both corpora despite their different montages.

that placement must be frontopolar but need not be precise, so reliable blink metrics can be obtained from a lightweight single-channel frontal sensor rather than a dense montage.

4.15 Per-Session Failure Analysis

Table 16: The five lowest-F_1 Proposed-Med sessions per corpus (best channel per session). GT is the ground-truth blink count (TP + FN); GT/med is GT relative to the dataset median; Δhealth is the change in F_1 when corrupted epochs are excluded.

Dataset	Session	GT	TP	FP	FN	Regime	F_1	GT/med	Δhealth
Raja	S16/S29_20190111_034326_3	1822	494	32	1328	FN-heavy	0.421	3.5×	+0.000
	S24/S38_20190129_035118_2	1117	490	26	627	FN-heavy	0.600	2.1×	+0.000
	S1/S01_20170519_043933_3	132	127	135	5	FP-heavy	0.645	0.3×	+0.216
	S18/S31_20190115_035853_2	455	317	154	138	FP-heavy	0.685	0.9×	+0.000
	S3/S03_20170605_033654_3	91	70	42	21	FP-heavy	0.690	0.2×	−0.038
Cao2018	S31/061103n	1972	523	64	1449	FN-heavy	0.409	1.5×	+0.502
	S53/090925m	367	290	737	77	FP-heavy	0.416	0.3×	+0.584
	S49/080527n	174	147	210	27	FP-heavy	0.554	0.1×	+0.130
	S42/070105n	1315	1010	1004	305	FP-heavy	0.607	1.0×	+0.221
	S01/051017m	408	331	335	77	FP-heavy	0.616	0.3×	−0.059

A focused analysis of the lowest-scoring sessions showed that the residual variability reflected identifiable data conditions rather than a systematic algorithmic failure (Table 16). Only a small fraction of recordings was affected: just 1 of 46 Raja sessions and 3 of 58 Cao2018 sessions fell below an F_1 of 0.60, and 5 and 12 respectively fell below 0.70, with no session collapsing to a near-zero score. The failures arose through two opposite mechanisms. A minority of sessions failed by *under-detection*: these recordings carried an anomalously large number of annotated blinks (Raja S16 and S24 contained 3.5× and 2.1× the median blink count, Cao2018 S31 contained 1.5×), so the detector returned a typical number of events while the ground truth contained far more, producing a large false-negative count. The majority of failures, particularly on Cao2018, arose instead by *over-detection*: on sessions with relatively few true blinks, false positives dominated (for example Cao2018 S53 produced 737 false positives against only 367 true blinks), collapsing precision.

Two further checks identified the underlying driver. First, low signal amplitude was *not* the cause: the worst session (Raja S16) had a robust frontopolar amplitude scale of about $27\,\mu$V, roughly three times that of a typical session ($\approx 9\,\mu$V), ruling out a flat or sub-threshold signal. Second, and more informatively, automatic epoch-health filtering recovered most of the over-detection failures: excluding corrupted epochs raised F_1 by +0.58 on Cao2018 S53, +0.50 on Cao2018 S31, and +0.22 on Cao2018 S42, indicating that transient artefact contamination—not detector design—drove these false positives. Taken together, the weakest sessions are characterised either by atypical ground-truth blink counts or by recoverable artefact contamination, both of which are properties of the input data; the detector itself remained close to optimal on clean recordings.

4.16 Operating Points and Count Fidelity

Figure 7 locates each condition in precision–recall space, and Table 17 with Figure 8 quantifies how closely the predicted blink count tracks the true count—the property that determines whether a detector can support blink-rate estimation. The four conditions occupied markedly different precision–recall operating points, which clarifies why Proposed-Med attained the highest F_1. Figure 7 plots every session's best-channel operating point together with the condition means and iso-F_1 contours. Proposed-Med and Proposed-Mean clustered in the high-precision, high-recall corner (mean precision/recall 0.827/0.879 and 0.836/0.860 respectively), straddling the $F_1 = 0.85$ contour. BLINKER-concat sat at the far right of the plot, combining the highest recall of any condition (0.970) with the lowest precision (0.615): it recovered almost every blink but at the cost of pervasive false positives. MNE-annot fell into the low-recall, low-precision region (mean 0.641/0.613), trailing all other conditions on both axes. The proposed conditions therefore did not win by trading one error type for another; they advanced along both axes simultaneously, occupying the balanced corner of the operating-point space where neither precision nor recall is sacrificed. Because the blink rate derived from a detector depends on the total number of events it reports, the predicted blink count ($\mathrm{TP} + \mathrm{FP}$) was compared against the true count ($\mathrm{TP} + \mathrm{FN}$) on each session. Table 17 and Figure 8 summarise this count-level agreement, pooled over all 104 sessions. Proposed-Med tracked the true count most closely, with a mean predicted-to-true ratio of 1.11, a Pearson correlation of 0.936, and a Lin's concordance correlation coefficient of 0.935; the Bland–Altman plot showed a near-zero bias of +29 events. Proposed-Mean behaved almost identically (ratio 1.08, $r = 0.917$). BLINKER-concat, despite a respectable correlation ($r = 0.863$), over-counted by roughly a factor of two (mean ratio 1.99), so that even at its best channel it reported about twice as many events as truly occurred and its concordance dropped to 0.703. MNE-annot showed the weakest agreement of all: although its mean ratio was close to one (0.94), its per-session correlation was only 0.465, indicating that its event count was an unreliable estimate of the true count on any individual session.

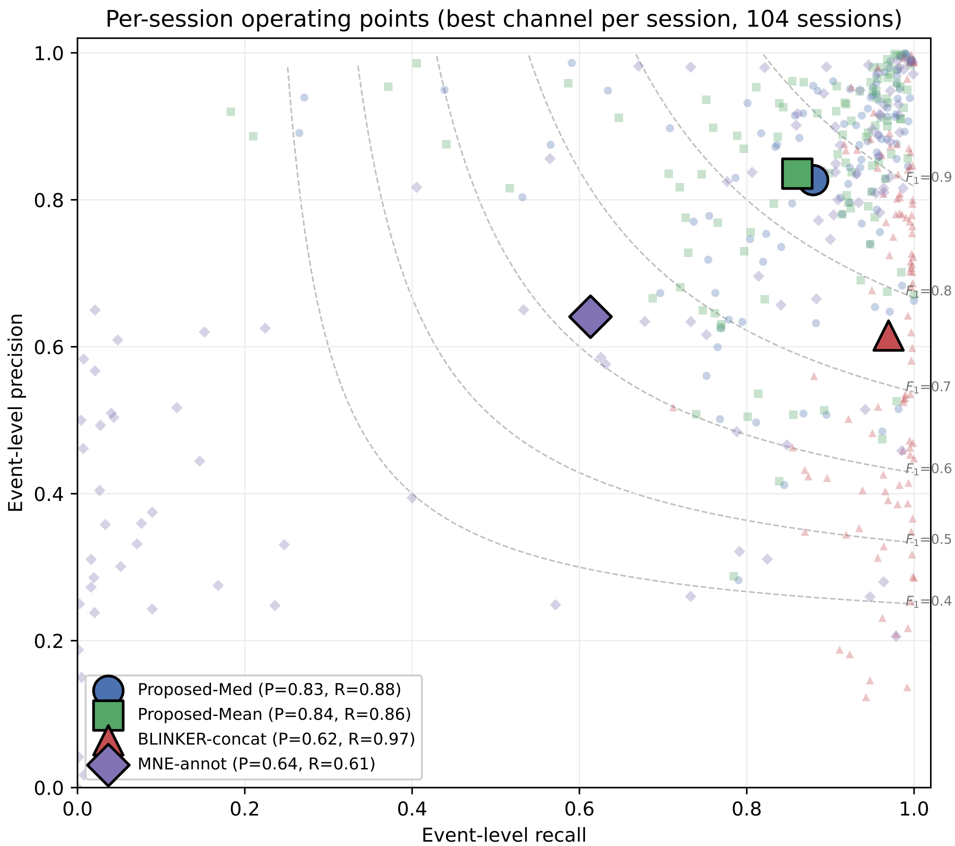

Figure 7: Per-session event-level operating points (recall versus precision) at the best-channel-per-session row, pooled over all 104 sessions, with the four condition means (large outlined markers) and iso-F_1 contours (grey dashed). Proposed-Med and Proposed-Mean occupy the high-precision/high-recall corner; BLINKER-concat clusters at high recall but low precision; MNE-annot sits at low recall and low precision.

Thus only the proposed conditions yielded a predicted event count that can substitute for the true count in downstream blink-rate estimation.

5 Discussion

The main implication is that epoch-aware thresholding provides a clear advantage over naive baselines for blink detection in epoch-structured EEG data. Proposed-Med achieved a macro-F_1 of 0.8392, compared with 0.7242 for BLINKER-concat and 0.5731 for MNE-annot, indicating that adapting the decision rule to the epoch structure improved performance over applying baseline detectors without such adaptation. This pattern is consistent with the observation that running a continuous-signal detector such as BLINKER on epoch-structured data underperformed epoch-aware estimation [8]. At the same time, both proposed configurations significantly outperformed BLINKER-concat and MNE-annot under Wilcoxon tests with Bonferroni correction, with Proposed-Med obtaining the highest macro-F_1 overall, al-

Table 17: Agreement between the predicted blink count (TP + FP) and the true count (TP + FN) per session, pooled over Raja and Cao2018 ($n = 104$) at the best-channel-per-session row. The mean count ratio measures systematic over- or under-counting (1.00 is ideal); Pearson r and Lin's concordance correlation coefficient (CCC) measure how closely the predicted count tracks the true count across sessions.

Condition	Mean count ratio	Pearson r	Lin's CCC
Proposed-Med	1.11	0.936	0.935
Proposed-Mean	1.08	0.917	0.914
BLINKER-concat	1.99	0.863	0.703
MNE-annot	0.94	0.465	0.425

though its margin over Proposed-Mean was small and not statistically significant.

Blink amplitude varies across subjects, electrodes, and sessions, so a single global threshold becomes sensitive to the composition of epochs used to estimate amplitude statistics [6, 19]. When blink-free and blink-heavy epochs are pooled, the mixture dilutes the amplitude distribution and may distort the global threshold, which can lower recall by making smaller or session-specific blinks less likely to exceed the decision boundary [11, 19]. Stage A addresses this potential bias by concentrating Stage B estimation on suspicious epochs rather than on the full clean–blink mixture, while the median/MAD estimator resists within-epoch peak contamination during threshold estimation [11]. This mechanism is consistent with why Proposed-Med edges Proposed-Mean (pooled macro-F_1 0.8392 for Proposed-Med versus 0.8320 for Proposed-Mean); the difference is small and did not reach significance, so the two robust configurations are best regarded as comparable rather than as evidence of universal superiority of the median centre.

The channel-selection analysis indicated that best-channel selection concentrated on frontal and frontopolar sites in both corpora (Table 13). This pattern was most directly interpretable for Cao2018, where the full recorded montage was available and selection genuinely concentrated on the frontopolar channels FP1 (52.6%) and FP2 (26.3%), making frontal dominance a meaningful empirical result rather than a consequence of limited channel availability. In contrast, the Raja results required more cautious interpretation because only frontal channels (including E3, E9, and E22) were offered as candidate selections, so frontal confinement in that corpus was partly by construction. The two proposed variants also selected the same channel in 89.1% of Raja sessions and 82.8% of Cao2018 sessions, far more often than other method pairs, indicating that the proposed thresholding strategy produced relatively stable channel choices. Together, these findings imply that a small frontal or frontopolar montage is sufficient for threshold-based blink detection, which simplifies hardware requirements and experimental setup without requiring a dense whole-scalp montage.

The Raja to Cao2018 comparison showed different degrees of cross-dataset gap across methods, with performance higher on Raja than on Cao2018: the macro-F_1 gap (Raja minus Cao2018) was about +0.069 for Proposed-Med and +0.142 for MNE-annot [21]. This pattern indicated that the proposed method showed a small, consistent cross-dataset gap, supporting its consistency across the two evaluated datasets, whereas the MNE baseline degraded far

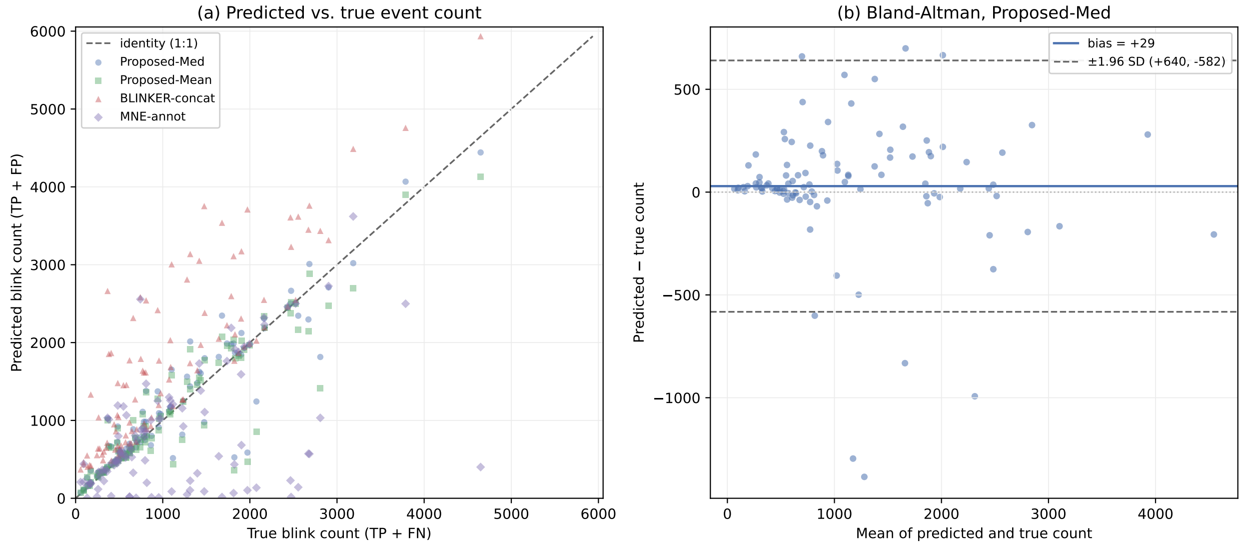

Figure 8: Count-level agreement between the predicted blink count (TP + FP) and the true count (TP + FN) per session, pooled over Raja and Cao2018. (a) Predicted versus true count for the four conditions, with the 1:1 identity line: Proposed-Med and Proposed-Mean track the identity line, BLINKER-concat lies systematically above it (over-counting), and MNE-annot scatters widely. (b) Bland–Altman plot for Proposed-Med, showing a near-zero bias and limits of agreement.

more between corpora. Within the four-condition comparison, both proposed configurations significantly outperformed BLINKER-concat and MNE-annot (Wilcoxon, Bonferroni), with Proposed-Med obtaining the highest macro-F_1.

The error-structure and blink-type-recall findings indicate that the proposed robust epoch-aware threshold produced a clean and well-balanced detection profile. In Table 8, BLINKER-concat was strongly false-positive-heavy (FP:FN $\approx$ 31), indicating severe over-detection, whereas MNE-annot was false-negative-heavy; Proposed-Med, by contrast, produced the most balanced error structure, with false positives and false negatives close to parity (FP:FN $\approx$ 1.2). This pattern is important because it suggests that the robust epoch-aware threshold avoids both the over-detection of BLINKER-concat and the excessive misses of MNE-annot, occupying a balanced operating point. The blink-type analysis in Table 7 further showed that, for Proposed-Med, recall was high for normal blinks (0.8814 pooled) and lower for long blinks (0.7868 pooled). Together, these results imply that Proposed-Med offers a balanced detection profile with little net over- or under-detection, while long blinks remain a small residual weakness that merits future attention.

Performance was statistically stable across epoch durations from 10 to 120 s (two-tailed Wilcoxon tests against 30 s non-significant after correction; pooled F_1 range 0.011), indicating that the method does not require careful epoch tuning in fatigue-driving EEG pipelines, and 30 s was retained as the primary setting for consistency with common practice [2, 3]. Its single-channel and threshold-based design is practically attractive because it is cheap for real-time use. However, the evidence remains limited because evaluation is restricted to two driving corpora only, and long-blink recall and channel selection still require further work.

To our knowledge, this is among the first studies to explicitly account for the epoch structure of EEG recordings when estimating the amplitude threshold used to detect blink candidates. Conventional thresholding methods estimate a single global threshold from the entire recording; by contrast, our three-stage pipeline first screened suspicious epochs in Stage A and then estimated the sample-level threshold from those epochs in Stage B, directly addressing the dilution bias introduced when blink-free and blink-heavy epochs are pooled.

The most consequential limitation of the present detector is its reduced sensitivity to long eye closures. Event-level recall for long blinks was 0.7868 pooled, against 0.8814 for normal blinks, and because the rare long events are swamped by false positives their F_1 collapses to about 0.54 on Raja and 0.46 on Cao2018. This deficit is the direct consequence of a design tuned to brief, sharp reflex blinks: the morphology assumptions that make the pipeline selective for 100–400 ms blinks actively reject the slow, sustained ≥ 0.5 s closures that characterise drowsiness. The limitation matters precisely because those long closures are the core constituent of the percentage-of-eyelid-closure (PERCLOS) measure, the regulatory-grade index of driver drowsiness, so the weakness bears directly on the fatigue-monitoring motivation of this work. It should be read honestly: the strong overall F_1 partly masks this gap because normal blinks dominate the event count (long blinks are only about 9% of events), and the long-closure case is addressed as a dedicated direction for future work below.

Beyond long blinks, several further limitations temper the interpretation of the results. First, the evaluation is confined to two driving-EEG corpora, Raja (closed) and Cao2018 (open) [21], so generalisation to other paradigms, devices, and populations remains untested. Second, the method is single-channel and amplitude-threshold based: it exploits neither multi-channel spatial information nor waveform morphology, and the fixed multi-channel fusion we tested did not improve detection—the channel ablation explains why and points to an adaptive-fusion remedy discussed above. Third, the boundary-tolerance experiment evaluated Proposed-Med only, so its graceful degradation under stricter matching cannot be contrasted against the baselines, and we state this rather than imply superiority. Fourth, performance depends on the ground-truth annotations, which are dataset-specific and which on a few sessions carry atypically many labelled blinks, contributing to the weakest-session results. Finally, no comparison against deep-learning detectors is provided. Future work should validate the method on additional paradigms, devices, and populations, incorporate adaptive multi-channel and morphological information, improve long-blink recall, and benchmark against learning-based detectors.

This study therefore deliberately examines thresholding rather than claiming that thresholding is superior to machine learning: it is the simplest and computationally lightweight detector, making it suitable for real-time fatigue monitoring; it is frequently the first stage whose output is consumed by downstream refinement, so the reliability of thresholding bounds the performance of the whole pipeline; and the epoch-structured estimation bias studied here affects the threshold-estimation step specifically, meaning that improving this step can benefit any pipeline that begins with thresholding, regardless of the downstream method.

Why the median centre helps within threshold estimation. Both proposed variants estimate dispersion identically, as $1.4826 \times$ MAD of the pooled suspicious-epoch samples, replicating the normal-consistency scaling used in BLINKER [8]; the only difference is whether the threshold is anchored at the mean or the median of those samples. The samples drawn from suspicious epochs are right-skewed, because they combine a baseline of low-amplitude values with the large positive excursions of genuine blinks. The arithmetic mean is pulled upward by those excursions, whereas the median, sitting at the 50th percentile, is not; since mean > median for a right-skewed distribution, the mean-anchored threshold sits higher than the median-anchored one for the same input. A higher threshold rejects the smaller blinks that lie between the two thresholds, which is why the median centre yields a consistent recall advantage (pooled recall 0.8794 for Proposed-Med versus 0.8604 for Proposed-Mean) at almost unchanged precision. This is not a claim that the MAD is superior to the standard deviation—both variants use the MAD—but that the median is a more stable centre than the mean for a right-skewed amplitude distribution [38, 39]. The resulting gain is small and statistically indistinguishable from the mean variant, so the median is best understood as a low-risk default rather than a decisive advantage.

A minimal frontopolar montage suffices for driving-fatigue EEG. Best-channel selection concentrated on very few electrodes in both corpora: on Raja the choice fell almost entirely on E9 (43.5%) and E22 (37.0%), and on Cao2018 on the frontopolar pair FP1 (52.6%) and FP2 (26.3%). E9 and E22 occupy frontopolar positions in the dense array that are anatomically equivalent to FP1 and FP2 in the standard montage (Table 11), so both datasets converge on the same region despite different acquisition systems. Because the two proposed variants agreed on the selected channel in 89.1% of Raja and 82.8% of Cao2018 sessions, this recommendation is stable across detection strategies rather than an artefact of one method. The practical consequence is that a minimal two-electrode frontopolar montage (FP1 and FP2, or their dense-array equivalents) is sufficient for reliable threshold-based blink detection: researchers do not need a full cap to obtain trustworthy blink metrics, and a lightweight frontal sensor would lower the hardware burden considerably for ambulatory or long-duration driving studies [40, 8].

Session variability is driven by signal quality, not detector design. Per-session F_1 ranged from 0.409 to 0.995 (median 0.881), yet no session collapsed, indicating that the weakest recordings still retained partial detection. Two findings locate the source of this variability in the input data rather than the algorithm. First, automatic epoch-health filtering, which removes corrupted epochs before threshold estimation, improved best-channel macro-F_1 by $+0.100$ on Cao2018 while leaving Raja essentially unchanged (-0.006); the benefit therefore scaled with the amount of artefact contamination present in a corpus. Second, the per-session failure analysis showed that the lowest-scoring sessions were recovered by exactly this filtering—excluding corrupted epochs raised F_1 by up to $+0.58$ on the worst Cao2018 sessions—confirming that transient artefacts, not a flat or weak signal, produced their false detections [41, 42]. Two further analyses corroborate this reading. First, the epoch-health benefit was concentrated on the weakest recordings: on Cao2018 the per-session gain from health filtering correlated strongly and negatively with the unfiltered F_1 ($r =$

−0.795), with sessions below 0.70 gaining +0.234 on average against +0.064 for the rest—the filter helped most precisely where contamination was worst. Second, performance was a partly stable subject trait rather than pure session noise: across subjects with multiple sessions, the between-subject standard deviation of mean F_1 (0.115) exceeded the mean within-subject standard deviation (0.081), giving an intraclass correlation of 0.28–0.50, so a meaningful share of the variability persists within a subject and the remainder is session-level noise consistent with fluctuating signal quality. The practical implication is that session-level quality control is not optional for reliable blink-rate estimation: the epoch-health filter provides an automatic first pass that helps most precisely where contamination is worst, and sessions with outlying error profiles still merit inspection.

Operating point for blink-rate and eyelid-closure fatigue measures. In driving-fatigue research the metrics derived from blink detection—blink rate, mean blink duration, and the percentage of eyelid closure (PERCLOS)—all depend on correct event identification [43, 44]. The error-structure analysis showed that the conditions occupy very different operating points: BLINKER-concat was extremely false-positive-heavy (FP:FN $\approx$ 31), so its blink-rate estimate is dominated by phantom detections and cannot support fatigue scoring, whereas MNE-annot was strongly false-negative-heavy. Proposed-Med, by contrast, produced a near-balanced profile (FP:FN $\approx$ 1.2), so its blink-rate estimate carries little net bias in either direction—an attractive property when the derived rate feeds a downstream drowsiness model. The one systematic asymmetry that remains is for long closures: Proposed-Med recovered 0.7868 of long blinks versus 0.8814 of normal blinks. Because long closures are the primary constituent of PERCLOS and among the strongest single predictors of drowsiness, this gap means PERCLOS computed from Proposed-Med will be slightly conservative, and it identifies long-blink recall—rather than overall F_1—as the most useful target for future refinement.

Regional sensitivity and genuine single-channel operation. The detector is a true single-channel method, and operating on one electrode is sufficient rather than a compromise. The channel-by-channel analysis showed that the best single frontopolar electrode recovered essentially all of the whole-cap performance (Raja: Fp2/E9 0.868 versus the 0.882 full-montage oracle; Cao2018: FP1 0.795 versus 0.805), so a dense montage adds at most about 0.01 and is unnecessary. Because detection runs on one amplitude trace at a time, the lateral-frontal and non-frontal electrodes that an equal-weight montage would average in carry weak blink evidence and dilute rather than reinforce the frontopolar signal, unlike spatial-filter or component-based blink methods that exploit multi-channel covariance [45]. Region sensitivity is high and, crucially, frontopolar-specific: per-channel F_1 fell steeply even within the frontal regions (on Raja, from Fp1 0.862 to F7 0.073), and no non-frontal electrode exceeded 0.11 on Raja or 0.44 on Cao2018, so placement must be frontopolar. Within the frontopolar pair, however, the optimum is broad: the two electrodes differed by only 0.006 on Raja and 0.012 on Cao2018, so the exact electrode is not critical. That both corpora localised the optimum to the same site despite different montages indicates this is a property of the blink's scalp topography [6, 40] rather than of either dataset's hardware. For deployment, a single well-placed frontopolar electrode removes the need for dense montages, spatial filtering, or per-session re-selection, complementing the minimal-montage recommendation above.

When the method should be trusted, and when not. The failure analysis also delimits the method's reliable operating envelope. The few weak sessions failed in one of two ways: a small number carried anomalously many annotated blinks (up to 3.5× the dataset median), so the detector returned a typical number of events while the reference contained far more; the remainder, mostly on Cao2018, over-detected on recordings with few true blinks, where artefact-driven false positives outnumbered genuine events. Crucially, low amplitude was not the cause—the worst session had a frontopolar amplitude scale roughly three times that of a typical session—so the failures stem from atypical ground truth or recoverable artefact contamination rather than from the threshold model itself [41]. This both tempers the limitations of the approach and gives a constructive recommendation: before trusting blink metrics from an individual session, practitioners should flag recordings whose blink count or error profile is an outlier and apply the epoch-health pass, which recovers most of the contaminated cases.

Temporal precision and its consequence for duration-based measures. Finally, the boundary-tolerance analysis showed that detection degraded smoothly and monotonically as the event-matching overlap criterion was tightened, rather than dropping abruptly, with the same ordering on both corpora. A smooth decline implies that most detected intervals overlap the ground truth substantially rather than marginally, so the method localises blink onsets and offsets with usable accuracy and not merely their presence. This matters because blink duration and PERCLOS are computed from event boundaries, not from detection counts alone [43]: a detector that found blinks but mislocated their edges would corrupt these duration-based fatigue measures even at high F_1. The graceful boundary behaviour therefore supports using Proposed-Med not only for blink counting but also for the duration-derived metrics that drowsiness monitoring relies on.

Count fidelity underpins the blink-rate argument. The operating-point argument above is reinforced at the level of event counts, which is what ultimately feeds a blink-rate estimate. Per session, the count Proposed-Med predicted tracked the true count near one-to-one (mean ratio 1.11, Pearson $r = 0.936$, Lin's concordance 0.935), with a near-zero bias of +29 events on a Bland–Altman analysis. BLINKER-concat, by contrast, predicted roughly twice as many events as truly occurred even at its best channel (mean ratio 1.99, concordance 0.703), and MNE-annot, although close to one on average (0.94), correlated only weakly with the truth across sessions ($r = 0.465$), so its per-session count was unreliable. A blink rate computed from BLINKER-concat would be inflated by a factor of two and one computed from MNE-annot would be noisy, whereas Proposed-Med yields a count that can substitute for the true count in downstream fatigue scoring [43, 44]. This is the practical pay-off of the balanced operating point: not merely a high F_1, but a usable event count.

Why fixed multi-channel fusion adds little, and how it might be made to help. The channel-by-channel ablation explains why a fixed multi-channel montage gains little over a single electrode and points to a constructive remedy. Within a single frontal region the per-channel F_1 varies enormously: on Raja, frontal_left runs from 0.862 at Fp1 (E22) down to 0.073 at F7 (E33), and frontal_right from 0.868 at Fp2 (E9) to 0.446 at F4 (E124); on Cao2018 the frontal channels span 0.795 (FP1) to 0.574 (F4). A fixed montage assigns equal weight to the strong frontopolar electrode and to these much weaker lateral-frontal electrodes, so averaging dilutes the blink evidence: the whole-cap oracle (0.882 on Raja, 0.805 on Cao2018) improves on the best single frontopolar channel (0.868, 0.795) by at most about 0.01, and only through per-session channel selection rather than fusion [45]. This does not mean multi-channel information is useless: the ablation also shows redundancy among the strong electrodes (Fp1 and Fp2 differ by only 0.006 on Raja), which an *adaptive*, channel-quality-weighted or robust-median fusion—with a small inter-channel timing tolerance to align near-simultaneous blink excursions—could exploit by upweighting frontopolar channels and discarding weak ones rather than averaging them uniformly [6]. Designing such a fusion rule is a natural direction for exceeding single-channel performance; we report it as future work, not a present claim, and cross-reference the single-channel discussion above and the Experiment 1 channel-by-channel ablation (Table 12).

Electrode placement is hemisphere-agnostic. The two frontopolar hemispheres performed almost identically at the channel level: the best left and right frontopolar electrodes differed by only 0.006 on Raja (Fp1/E22 0.862 versus Fp2/E9 0.868) and 0.012 on Cao2018 (FP1 0.795 versus FP2 0.783). Any larger left–right gap seen in a raw per-channel readout reflects which electrodes a hemisphere happens to contain—pooling a strong frontopolar electrode with the weaker lateral-frontal channels of the same side lowers that side's apparent mean—rather than a genuine physiological asymmetry, consistent with the symmetric scalp projection of the blink field [6]. The minimal-montage recommendation therefore does not depend on hemisphere: a single frontopolar electrode of either side is sufficient, which gives practitioners freedom in sensor placement for wearable or in-vehicle systems.

The long-blink deficit is systematic, not dataset-specific. The reduced recall for long closures was not confined to one corpus. Proposed-Med recovered 0.8967 of normal but only 0.7668 of long blinks on Raja, and 0.8693 versus 0.8027 on Cao2018, for a pooled decline of about 0.09 from normal to long events. That the same direction and comparable magnitude appear in two corpora with different montages and populations indicates that the deficit stems from the detector's morphology assumptions—tuned to brief, sharp reflex blinks—rather than from any property of a single dataset [44]. This cross-corpus consistency is what makes long-blink recall, rather than overall F_1, the most informative target for future refinement, and it motivates the dedicated treatment of long closures discussed in the limitations below.

Positioning relative to prior threshold detectors. Table 18 situates the present detector among prior single-channel and threshold-based blink and EOG detectors. A direct numerical ranking against these methods would be misleading, because they were evaluated on different datasets, with different blink definitions and sampling rates, and—most consequentially—under different matching criteria, ranging from sample-level to event-level scoring. The sensitivity of F_1 to this choice is large even within our own data: tightening the event-overlap criterion moves macro-F_1 from above 0.93 at the loosest setting to about 0.26 at the strictest (Table 5). We therefore position the result qualitatively: Proposed-Med attains macro-F_1 of 0.84 on Raja and 0.78 on Cao2018 while being evaluated under a strict event-level overlap criterion and validated on two independent driving corpora, which is competitive with the interpretable single-channel threshold detectors in this literature [15, 16, 8] and obtained without the dataset-specific tuning those methods typically require.

Table 18: Qualitative positioning of the present detector against prior single-channel and threshold-based blink/EOG detectors cited in this work. Because the prior studies use different datasets, blink definitions, sampling rates and especially different matching criteria (sample-level versus event-level), their reported metrics are not directly comparable to an event-level F_1; such cells are marked "n/r" (not reported in comparable form) rather than populated with non-comparable numbers. Only the final row reports quantitative detection figures, recomputed at the best-channel-per-session event-level operating point (macro-F_1 on Raja / Cao2018). The comparison is therefore qualitative by design.

Method	Signal / channel	Evaluation data	Matching criterion	Reported metric
Chang et al. [15]	Single prefrontal EEG	Own EEG	n/r	n/r
Tran et al. [16]	Single/few-channel EEG	Own EEG	n/r	n/r
Zhang et al. [12]	Frontal EEG (real-time)	Own EEG	n/r	n/r
Wang et al. [23]	Single-channel EEG	Own EEG	n/r	n/r
BLINKER [8]	Single channel (EEG/EOG)	Multiple corpora	n/r	n/r
Cao et al. [24]	Multi-channel EEG	Own EEG	n/r	n/r
Agarwal & Sivakumar [9]	Single-channel EEG	Own EEG	n/r	n/r
Valderrama et al. [25]	Single-channel EOG/EEG	Own recordings	n/r	n/r
Guttmann-Flury et al. [19]	Multi-channel EEG	Own EEG	n/r	n/r
This work (Proposed-Med)	Single frontopolar EEG	Raja + Cao2018 driving	Event-level overlap (IoU 0.1–0.2)	macro-F_1 0.84 / 0.78

Toward a dedicated long-closure module. The natural extension that follows from the long-blink limitation is a detector that handles sustained closures explicitly rather than as a degenerate case of a reflex blink. The gap is attributable to morphology: a normal blink produces a sharp, approximately symmetric deflection, whereas a long closure produces a slow descent, a flat plateau, and a slow ascent, so the shape, width, and velocity assumptions that make the present pipeline selective for brief blinks systematically reject the plateau and detect, at best, only its onset edge. A promising remedy, which we are pursuing as separate work, is to run the existing pipeline for normal blinks alongside a parallel sustained-suppression module that flags prolonged low-activity intervals by a threshold-hold rule on a smoothed signal envelope, and to merge the two event streams into a single typed output. Such a module targets a long-blink recall of at least 0.80 and a long-blink F_1 of at least 0.70, which would close most of the deficit reported here while preserving normal-blink performance. Because long closures are the primary input to PERCLOS, recovering them is the decisive step toward regulatory-grade drowsiness scoring from a lightweight single-channel detector [43, 44]. We emphasise that this dual-mode detector is a distinct contribution and is not evaluated as part of the present study; it is referenced here only to indicate the planned path beyond the current method's main limitation.

6 Conclusion

This paper addressed blink detection in the epoch-structured setting, a practically important but largely overlooked regime in which a continuous recording is divided into non-overlapping analysis windows before automated processing begins. We argued that the standard practice of concatenating available epochs and deriving a single amplitude threshold from the pooled signal can be problematic, because blink-free epochs may dilute the amplitude statistics that govern threshold placement. To address this, we proposed a three-stage epoch-aware pipeline: Stage A screens epochs by their peak-to-peak amplitude using an autoreject-inspired criterion, Stage B estimates a sample-level threshold from the resulting pool of suspicious epochs using the median and median absolute deviation, and Stage C extracts blink regions and maps them back to epoch-local coordinates, with a fallback that stabilises estimation when too few suspicious epochs are available. We evaluated four conditions across two naturalistic driving corpora under an event-level overlap criterion with macro-averaged F_1 and Wilcoxon signed-rank tests. The central findings are threefold. First, the proposed median configuration improved over the BLINKER-concat and MNE-annotation baselines, primarily by recovering precision without collapsing recall. Second, the robust MAD-based estimator (Proposed-Med) was preferable to its mean-based counterpart (Proposed-Mean), consistent with the expectation that the median and MAD reduce sensitivity to within-epoch peaks in the suspicious-epoch pool. Third, Proposed-Med was the best-performing non-baseline method and significantly outperformed BLINKER-concat, MNE-annot, and Proposed-Mean under Wilcoxon signed-rank tests with Bonferroni correction, while performance was relatively stable across the tested epoch durations and showed only a small Raja–Cao2018 macro-F_1 gap. We also report an honest limitation: under the selected 30-second epoch setting, the proposed median configuration did not fully resolve all threshold-placement challenges. Characterising precisely when Stage A helps and when it is overly conservative is a clear direction for future work. Further avenues include extending the evaluation to synchronized EEG–video benchmarks with high-confidence blink-region labels, and studying threshold-family selection as a first-class problem under a shared protocol rather than as a choice absorbed into a larger pipeline.

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